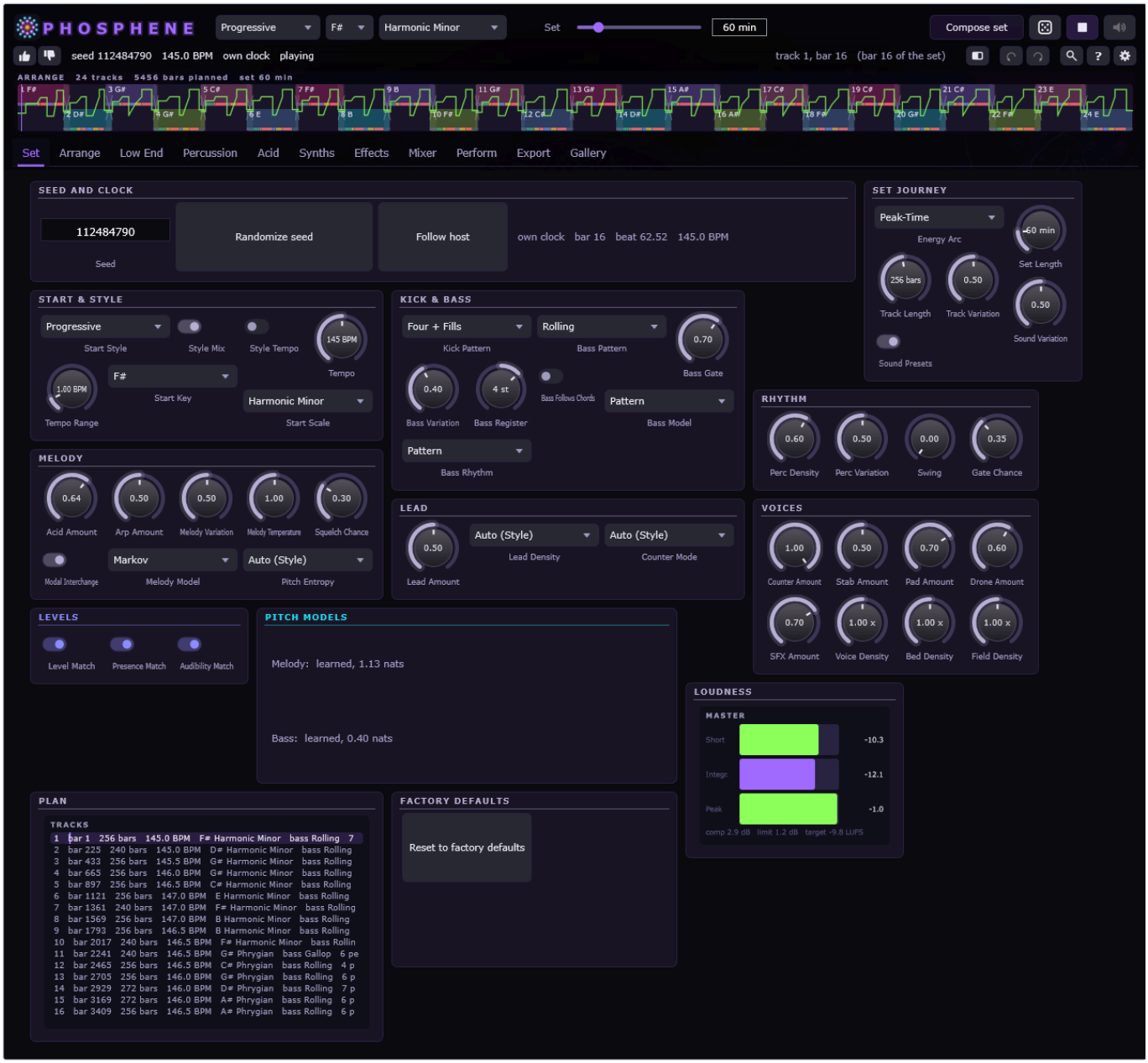
# Phosphene

Manual · version 1.5.0




1556 parameters · 18 tabs · 4 perform macros · every picture in this manual is the plugin drawing itself at design size

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## 1. About

Phosphene composes and plays complete psytrance sets. You give it a seed, a style and an energy arc; it writes the tracks -- their keys, tempi, forms, patterns and sounds -- and synthesises them in real time. There is no loop pool and no sampled instrument: every note is decided by the composer, and every sound is made by the engine -- except the spoken phrases and, on the Field track, field recordings of real places (and of NASA's missions), which a place cannot be synthesised without.

Two ideas run through the whole instrument. The first is that the same seed always gives the same set: bar 3000 is bar 3000 whether the night was played from the start or joined in the middle, which is what makes a set something you can save in seven lines of text and hand to someone else. The second is that the composer never writes a knob. For every track it chooses a sound preset for each synth and plays it, and on top of that it rides offsets -- a section opening a filter, the level match evening out the parts. The knobs stay yours: the tab shows the preset the composer plays, a thin bright ring shows where a value plays away from its knob, and the moment you turn a knob the synth is yours again (Own Sound).

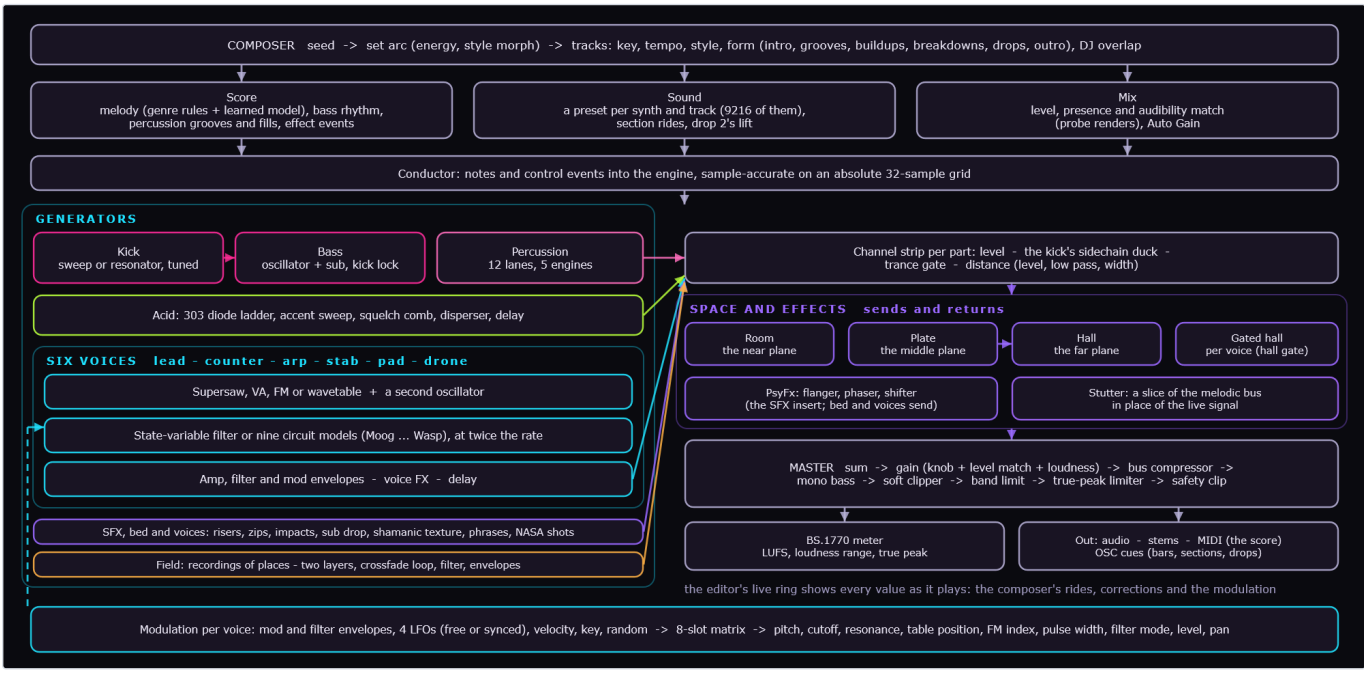
The panel has a tab per generator, plus the pages that are about the set rather than the sound: Set, Arrange, Mixer / Master, Perform and Gallery. Every control on every page is generated from the engine's own parameter tables, and so is this manual.

## 2. Signal flow

The picture is the whole instrument in the order it works. At the top the composer: a seed becomes a set arc, the arc becomes tracks with keys, tempi, styles and forms, and each track becomes a score, a choice of sounds and a mix -- the last measured by rendering parts of the track before it plays. The conductor hands notes and control events to the engine on an absolute grid of 32 samples, which is why a bar sounds the same however a host cuts its blocks.

On the left the generators, each in the colour of its tab: kick, bass and percussion, the acid, the six melodic voices with their oscillators, filter models and envelopes, the effects with the bed and the spoken voices, and the Field track's sampler. Every part has a channel strip -- level, the kick's sidechain duck, the trance gate and its distance -- and sends into three planes of space: a room near, a plate in the middle, a hall far behind, plus a gated hall a voice can choose. The master sums it all and brings it to loudness: gain, bus compressor, mono bass, soft clipper, band limit, a true-peak limiter and a safety clip, metered to BS.1770. Out go the audio, the stems, the score as MIDI and the cues over OSC.

Along the bottom the modulation of every voice: its envelopes, four LFOs and an eight-slot matrix.




The signal flow, drawn by the plugin (the same picture as the help page's).

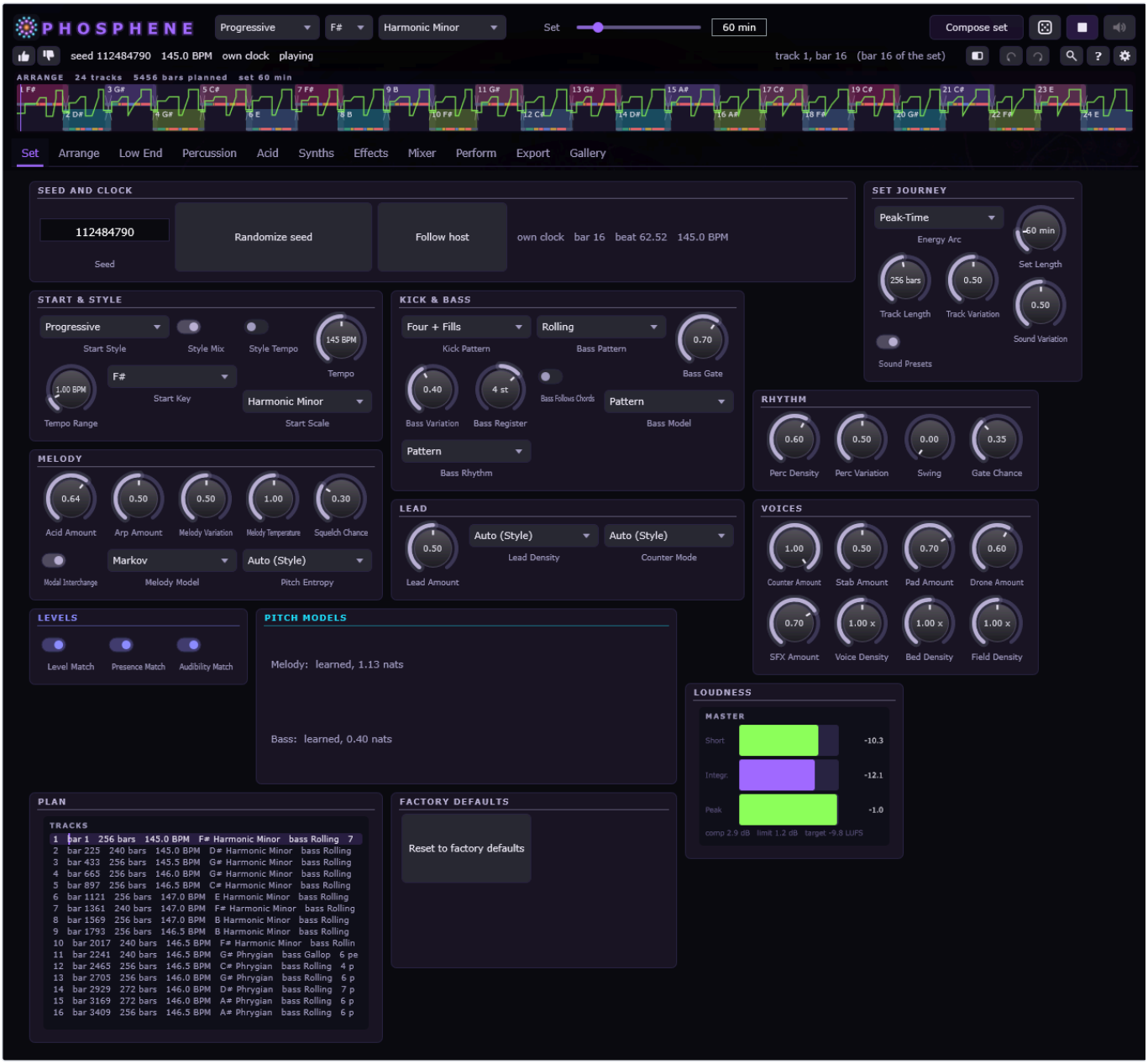
## 3. Set

The Set tab is where a night begins. The seed names the set; Randomize seed draws another one. Play and Stop run Phosphene's own clock, and Follow host hands the transport to the DAW instead -- under a host the tempo knob follows the host's tempo and the play head follows its position.

Planning a track measures it by rendering parts of it -- its loudness through the master, the brightness of its lines, how audible each line is against the lead. For the first track of a set that happens behind the music: it starts playing at once, its intro runs, and the measured levels ride in over a phrase long before the kick and the lines arrive. A jump to a track that has not been planned yet still waits for its measurements; the header says "planning" rather than going quiet without explanation. In the standalone this has usually happened before you press anything, because the composer thread starts working when the window opens.

The composer's own knobs are here, grouped by what they decide. Start & Style is where the night begins: the style profile, whether the style walks through the set (Style Mix) and whether each track takes its style's own tempo (Style Tempo, which then replaces Tempo and Tempo Range), and the start key and scale. They are called "start" because they are exactly that -- the first track plays them, and from the second on the key, the tempo and (with Style Mix) the style move from track to track as a DJ would move them. Set Journey is the shape of the night: the energy arc, the set length, the track length and how much tracks and sounds vary. Kick & Bass, Rhythm, Melody, Lead and Voices say how much of each part there is and how it is drawn; Levels holds the three automatic balances -- the loudness of whole tracks, the presence of the melodic lines, and the audibility of the quiet lines against the lead.

Export MIDI writes the score of the first N bars as a Standard MIDI File with a named track per part, section markers and the tempo map. The disk on the Export page writes a .phosset -- the seed, the style, the arc, the knobs that differ from their defaults, and every lock and reroll -- which phos_render will play back byte for byte.




The Set tab.

### START & STYLE

| Control | Key | Range | Default |
|---|---|---|---|
| Start Style | `compose.style` | Goa \| Full-On \| Progressive \| Dark Forest \| Hi-Tech | Full-On |

| | | | |
|---|---|---|---|
| Style Mix | `compose.style_mix` | off / on | on |
| Style Tempo | `compose.style_tempo` | off / on | off |
| Tempo | `compose.bpm` | 100 .. 190 BPM | 145 |
| Tempo Range | `compose.tempo_range` | 0 .. 10 BPM | 1 |
| Start Key | `compose.key` | C \| C# \| D \| D# \| E \| F \| F# \| G \| G# \| A \| A# \| B | F# |
| Start Scale | `compose.scale` | Aeolian \| Phrygian \| Harmonic Minor \| Phrygian Dominant \| Double Harmonic \| Dorian | Phrygian |

### SET JOURNEY

| Control | Key | Range | Default |
|---|---|---|---|
| Energy Arc | `compose.arc` | Warm-up \| Peak-Time \| Morning \| Closing \| Flat | Flat |
| Set Length | `compose.set_minutes` | 10 .. 300 min | 60 |
| Track Length | `compose.track_bars` | 32 .. 512 bars | 256 |
| Track Variation | `compose.track_variation` | 0 .. 1 | 0.5 |
| Sound Variation | `compose.sound_variation` | 0 .. 1 | 0.5 |
| Sound Presets | `compose.sound_presets` | off / on | on |

### KICK & BASS

| Control | Key | Range | Default |
|---|---|---|---|
| Kick Pattern | `compose.kick_pattern` | Four \| Four + Fills \| Off | Four + Fills |
| Bass Pattern | `compose.bass_pattern` | Rolling \| Gallop \| Skip \| Offbeat \| Triplet | Rolling |
| Bass Gate | `compose.bass_gate` | 0.2 .. 1 | 0.7 |
| Bass Variation | `compose.bass_variation` | 0 .. 1 | 0.4 |
| Bass Register | `compose.bass_register` | -12 .. 12 st | 0 |
| Bass Follows Chords | `compose.bass_follows_chords` | off / on | off |
| Bass Model | `compose.bass_model` | Pattern \| Neural | Pattern |
| Bass Rhythm | `compose.bass_rhythm` | Pattern \| Corpus | Pattern |

### RHYTHM

| Control | Key | Range | Default |
|---|---|---|---|
| Perc Density | `compose.perc_density` | 0 .. 1 | 0.6 |
| Perc Variation | `compose.perc_variation` | 0 .. 1 | 0.5 |
| Swing | `compose.swing` | 0 .. 0.3 | 0 |
| Gate Chance | `compose.gate_chance` | 0 .. 1 | 0.35 |

### MELODY

| Control | Key | Range | Default |
|---|---|---|---|
| Acid Amount | `compose.acid_amount` | 0 .. 1 | 0.6 |
| Arp Amount | `compose.arp_amount` | 0 .. 1 | 0.5 |
| Melody Variation | `compose.melody_variation` | 0 .. 1 | 0.5 |
| Melody Temperature | `compose.melody_temperature` | 0.5 .. 2 | 1 |
| Squelch Chance | `compose.squelch_chance` | 0 .. 1 | 0.3 |
| Modal Interchange | `compose.modal_interchange` | off / on | on |
| Melody Model | `compose.melody_model` | Markov \| Neural | Markov |
| Pitch Entropy | `compose.pitch_entropy` | Auto (Style) \| Low \| Medium \| High | Auto (Style) |

### LEAD

| Control | Key | Range | Default |
|---|---|---|---|
| Lead Amount | `compose.lead_amount` | 0 .. 1 | 0.5 |
| Lead Density | `compose.lead_density` | Auto (Style) \| Sparse \| Medium \| Dense | Auto (Style) |
| Counter Mode | `compose.counter_mode` | Auto (Style) \| Echo \| Answer \| Timbral \| Hocket | Auto (Style) |

### VOICES

| Control | Key | Range | Default |
|---|---|---|---|
| Counter Amount | `compose.counter_amount` | 0 .. 1 | 1 |
| Stab Amount | `compose.stab_amount` | 0 .. 1 | 0.5 |
| Pad Amount | `compose.pad_amount` | 0 .. 1 | 0.7 |
| Drone Amount | `compose.drone_amount` | 0 .. 1 | 0.6 |
| SFX Amount | `compose.sfx_amount` | 0 .. 1 | 0.7 |

| | | | |
|---|---|---|---|
| Voice Density | `compose.voice_density` | 0 .. 2 x | 1 |
| Bed Density | `compose.bed_density` | 0 .. 2 x | 1 |
| Field Density | `compose.field_density` | 0 .. 2 x | 1 |

### LEVELS

| Control | Key | Range | Default |
|---|---|---|---|
| Level Match | `compose.level_match` | off / on | on |
| Presence Match | `compose.presence_match` | off / on | on |
| Audibility Match | `compose.audibility_match` | off / on | on |

## 4. Arrange

The Arrange tab is the whole planned set on one page, and the curation loop that goes with it.

The strip across the top is the night: every planned track a block in the order it plays, the energy arc drawn over it, the play head in it. Underneath, each track gets a row of its own in which its own form fills the full width, so a section stays readable no matter how long the set is. The colours are the section categories: intro and outro grey, groove green, buildup amber, the pre-drop break violet, the drop red, the breakdown blue. Clicking any block jumps there.

The two small glyphs on every track and every section are the answer to the oldest complaint about generators, which is that they are boring. The padlock freezes that unit on the seed it has; the die draws it again. Because a locked unit keeps its old seed and nothing later reads anything an earlier one decided, rerolling one track leaves every other track bit-identical -- so you can keep what you like and reroll what you do not, one section at a time, until the set is yours. Reroll the set draws the whole journey again and leaves the locked tracks exactly where they were.




The Arrange tab.

## 5. Kick

The kick is the floor of the genre and the first thing to get right. Two engines: a swept sine whose pitch falls from Pitch Start to Pitch End with its own decay, and a damped phasor resonator with the retrigger behaviour of a TR-808. Punch is a second, much faster pitch envelope on top of the first; Click adds the attack transient; Drive and Clip decide how much of the body survives.

Two of its numbers are not free. The tail is limited so that the kick has fallen far enough by the time the first bass note starts, and the phase lock (on the Bass tab) can trim the kick's decay so that the kick and the bass fundamental meet in phase. Both are applied in the engine, after every knob and every variation.

The Kick Scope renders one hit with the values that play right now (the knobs plus the track's recipe; stopped, the knobs): its waveform, its pitch sweep on a log axis, its spectrum, and in figures where the sweep starts, the note it lands on (with its cents off it), its length to -60 dB and its peak.




The Kick tab.

### SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Kick Own Sound | `mix.kick_own` | off / on | off |

### PITCH

| Control | Key | Range | Default |
|---|---|---|---|
| Engine | `kick.engine` | Sweep \| Resonant | Sweep |
| Tune | `kick.tune` | Free \| Key | Key |
| Pitch Start | `kick.pitch_start` | 60 .. 1500 Hz | 330 |
| Pitch End | `kick.pitch_end` | 30 .. 120 Hz | 50 |
| Body Decay | `kick.pitch_decay` | 5 .. 150 ms | 13 |

### PUNCH & BODY

| Control | Key | Range | Default |
|---|---|---|---|
| Punch | `kick.punch` | 0 .. 1 | 0.5 |
| Punch Decay | `kick.punch_decay` | 0.5 .. 20 ms | 4 |
| Decay | `kick.amp_decay` | 20 .. 1500 ms | 240 |
| Attack | `kick.amp_attack` | 0 .. 10 ms | 0.2 |
| Hold | `kick.amp_hold` | 0 .. 150 ms | 12 |

### CLICK

| Control | Key | Range | Default |
|---|---|---|---|
| Click | `kick.click_level` | 0 .. 1 | 0.5 |
| Click Tone | `kick.click_tone` | 500 .. 12000 Hz | 4000 |
| Click Decay | `kick.click_decay` | 0.5 .. 30 ms | 3 |

### DRIVE & OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `kick.level` | -36 .. 6 dB | -6 |
| Drive | `kick.drive` | 0 .. 1 | 0.3 |
| Clip | `kick.clip` | Tanh \| Hard | Tanh |
| Tone | `kick.tone` | 200 .. 20000 Hz | 9000 |
| Tail Limit | `kick.tail_limit` | -60 .. 0 dB | -15 |

## 6. Bass

The rolling bass and the kick are one instrument with two halves. The oscillator is a PolyBLEP saw and pulse with a sine sub on the fundamental -- on the fundamental, not an octave down, because the octave-down sub fought the kick -- and the filter is a non-linear zero-delay ladder.

Kick Lock decides which of the two gives way: Bass follows kick sets the bass start phase to the kick's phase at that instant, Kick follows bass trims the kick's decay by bisection until the two agree. Off leaves them where they fall, which at 145 BPM means anything between -157 and +165 degrees.

Pattern and Gate are the note side: which of the five bass patterns the track prefers, and how long a note is as a fraction of its slot. The gate is never allowed to run shorter than half a period of the fundamental.

The bass is a pedal on the root. Its figures at the phrase ends may touch the b2 or the leading tone only as a pickup -- at most two sixteenths, at the end of the bar, into the next downbeat; anywhere else the figure takes the fifth, because under the drone's tonic those notes are a minor ninth in the low register. Bass Follows Chords (Set tab) moves the bass to the root the pad holds, and only where the pad plays; the drone moves with it.

The Bass Scope renders one eighth note of the bass on the key's root with the values that play right now (the knobs plus the track's recipe; stopped, the knobs): its waveform, the filter's cutoff over the note as the synth computes it, its spectrum, and the note and the filter's travel in figures.




The Bass tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Bass Own Sound | `mix.bass_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Wave | `bass.wave` | 0 .. 1 | 0.15 |
| Sub Mode | `bass.sub_mode` | Mixed \| Split | Split |

| | | | |
|---|---|---|---|
| Pulse Width | `bass.pulse_width` | 0.05 .. 0.95 | 0.5 |
| Sub | `bass.sub` | 0 .. 1 | 0.3 |
| Sub Octave | `bass.sub_octave` | 0 .. 1 | 0.5 |
| Split | `bass.split_ratio` | 1.2 .. 3 x f0 | 2 |
| Start Phase | `bass.start_phase` | 0 .. 1 | 0.5 |
| Retrigger | `bass.retrigger` | off / on | on |
| Kick Lock | `bass.kick_lock` | Off \| Bass follows kick \| Kick follows bass | Kick follows bass |

### FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `bass.filter_model` | Bass Ladder \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | Bass Ladder |
| Cutoff | `bass.cutoff` | 20 .. 10000 Hz | 240 |
| Resonance | `bass.resonance` | 0 .. 1 | 0.3 |
| Env Amount | `bass.env_amount` | 0 .. 8 oct | 4 |
| Filter Mode | `bass.filter_mode` | 0 .. 1 | 0 |
| Key Track | `bass.key_track` | 0 .. 1 | 0.6 |
| Vel > Cutoff | `bass.vel_to_cutoff` | 0 .. 1 | 0.25 |

### FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `bass.filt_attack` | 0.1 .. 4000 ms | 0.1 |
| Filter Decay | `bass.filter_decay` | 3 .. 1000 ms | 75 |
| Filter Sustain | `bass.filt_sustain` | 0 .. 1 | 0 |
| Filter Release | `bass.filt_release` | 5 .. 8000 ms | 300 |

### BITE

| Control | Key | Range | Default |
|---|---|---|---|
| Bite | `bass.bite` | 0 .. 1 | 0.5 |
| Bite Cutoff | `bass.bite_cutoff` | 100 .. 4000 Hz | 500 |
| Bite Reso | `bass.bite_resonance` | 0 .. 1 | 0.2 |
| Bite Env | `bass.bite_env` | 0 .. 5 oct | 3 |
| Bite Decay | `bass.bite_decay` | 5 .. 1000 ms | 90 |
| Bite Drive | `bass.bite_drive` | 0 .. 1 | 0.5 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Drive | `bass.drive` | 0 .. 1 | 0.3 |
| Attack | `bass.amp_attack` | 0.1 .. 50 ms | 0.8 |
| Decay | `bass.amp_decay` | 5 .. 2000 ms | 180 |
| Sustain | `bass.amp_sustain` | 0 .. 1 | 0.55 |
| Release | `bass.amp_release` | 1 .. 500 ms | 10 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `bass.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `bass.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `bass.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `bass.menv_release` | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `bass.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `bass.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `bass.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `bass.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `bass.lfo1_fade` | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `bass.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `bass.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `bass.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `bass.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `bass.lfo2_fade` | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `bass.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `bass.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `bass.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `bass.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `bass.lfo3_fade` | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `bass.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `bass.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `bass.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `bass.lfo4_retrig` | off / on | off |
| LFO 4 Fade | `bass.lfo4_fade` | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | `bass.mx1_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | `bass.mx1_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 1 Amount | `bass.mx1_amount` | -1 .. 1 | 0 |
| Mod 2 Source | `bass.mx2_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | `bass.mx2_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 2 Amount | `bass.mx2_amount` | -1 .. 1 | 0 |
| Mod 3 Source | `bass.mx3_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | `bass.mx3_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 3 Amount | `bass.mx3_amount` | -1 .. 1 | 0 |
| Mod 4 Source | `bass.mx4_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | `bass.mx4_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 4 Amount | `bass.mx4_amount` | -1 .. 1 | 0 |
| Mod 5 Source | `bass.mx5_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | `bass.mx5_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 5 Amount | `bass.mx5_amount` | -1 .. 1 | 0 |
| Mod 6 Source | `bass.mx6_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | `bass.mx6_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 6 Amount | `bass.mx6_amount` | -1 .. 1 | 0 |
| Mod 7 Source | `bass.mx7_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | `bass.mx7_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 7 Amount | `bass.mx7_amount` | -1 .. 1 | 0 |
| Mod 8 Source | `bass.mx8_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | `bass.mx8_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 8 Amount | `bass.mx8_amount` | -1 .. 1 | 0 |

### DUCK & LEVEL

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `bass.level` | -36 .. 6 dB | -3 |
| Duck Depth | `bass.duck_depth` | 0 .. 1 | 0.37 |
| Duck Hold | `bass.duck_hold` | 0 .. 200 ms | 25 |
| Duck Release | `bass.duck_release` | 5 .. 500 ms | 60 |

## 7. Percussion

Twelve lanes, each a complete voice: a tone with its own pitch envelope and FM, four modal sets, the 808 square stack, noise with clap bursts, a filter, a low cut, drive, choke and panning. The role of a lane decides what it plays in the groove and which General MIDI note it exports as.

The patterns are euclidean, rotated to a middle amount of syncopation rather than to none, and the decisions are made per four-bar phrase. Every lane is high-passed at 150 Hz at the least: the bottom of the mix belongs to the kick and the bass, and a tom that reaches into it takes the floor away.

On a metal lane, Noise trades against the square stack: at 1 the squares keep 20 % of their level. Six summed squares are six steady tones, and a cymbal made mostly of them rings like feedback. The default hats and ride therefore run almost all noise, with their low cuts at 3 .. 3.5 kHz.




The Percussion tab.

The 12 lanes of the kit share one table; lane 1 is printed here and the others are the same controls under the keys `perc2.*` to `perc12.*`.

### LANE

| Control | Key | Range | Default |
|---|---|---|---|
| Active | `perc1.active` | off / on | on |
| Role | `perc1.role` | Closed Hat \| Open Hat \| Ride \| Crash \| Clap \| Snare \| Rim \| Shaker \| Tom \| Conga \| Zap \| Blip | Closed Hat |
| Engine | `perc1.engine` | Noise \| Metal \| Modal \| Tone \| FM | Metal |
| Density | `perc1.density` | 0 .. 1 | 0.6 |
| Choke Group | `perc1.choke` | 0 .. 4 | 1 |
| Cut Tracks Pitch | `perc1.cut_track` | 0 .. 2 | 0 |

### SOURCE

| Control | Key | Range | Default |
|---|---|---|---|
| Pitch | `perc1.pitch` | 40 .. 12000 Hz | 400 |
| Pitch Amount | `perc1.pitch_amount` | 1 .. 16 x | 1 |
| Pitch Decay | `perc1.pitch_decay` | 0.5 .. 300 ms | 10 |
| Tune to Key | `perc1.tune` | off / on | off |
| Shift | `perc1.shift` | -10 .. 10 ms | 0 |
| FM Ratio | `perc1.fm_ratio` | 0.25 .. 8 | 1.41 |
| FM Index | `perc1.fm_index` | 0 .. 8 | 0 |

| | | | |
|---|---|---|---|
| Modes | `perc1.mode_set` | Membrane \| Bar \| Harmonic | Membrane |
| Mode Damping | `perc1.mode_damp` | 0 .. 1 | 0.5 |
| Metal Scale | `perc1.metal_scale` | 0.25 .. 4 x | 1 |

### NOISE

| Control | Key | Range | Default |
|---|---|---|---|
| Noise | `perc1.noise` | 0 .. 1 | 1 |
| Noise Decay | `perc1.noise_decay` | 2 .. 3000 ms | 35 |
| Bursts | `perc1.bursts` | 1 .. 6 | 1 |
| Burst Spacing | `perc1.burst_spacing` | 2 .. 40 ms | 10 |

### FILTER & SHAPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter | `perc1.filter` | Low Pass \| Band Pass \| High Pass | Low Pass |
| Cutoff | `perc1.cutoff` | 100 .. 18000 Hz | 12000 |
| Resonance | `perc1.resonance` | 0 .. 1 | 0.05 |
| Low Cut | `perc1.low_cut` | 150 .. 8000 Hz | 3000 |
| Decay | `perc1.decay` | 2 .. 3000 ms | 45 |
| Drive | `perc1.drive` | 0 .. 1 | 0 |

### OUTPUT & MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `perc1.level` | -36 .. 6 dB | 1.5 |
| Pan | `perc1.pan` | -1 .. 1 | 0.45 |
| Pan Depth | `perc1.pan_depth` | 0 .. 1 | 1 |
| Pan Period | `perc1.pan_bars` | 0.0625 .. 16 bars | 0.1875 |

## 8. Acid

A diode ladder after Zavalishin, solved from the inside out, with an accent that charges a sweep capacitor the way the original did -- three accents in a row open further than one. Slide is an exponential glide into the next note, Squelch a cutoff start value plus a comb tuned to the note's period. The tempo delay is in beats, so it follows the set.

In the Phrygian modes the b2 is a degree of the acid's own: the F of an E - F - E riff, on any step and held too. The other colour tones -- the upper note of an augmented second -- stay short neighbours that resolve to the tonic.




The Acid tab.

### SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Acid Own Sound | `mix.acid_own` | off / on | off |

### VOICE

| Control | Key | Range | Default |
|---|---|---|---|
| Wave | `acid.wave` | 0 .. 1 | 0 |
| Slide Time | `acid.slide_time` | 5 .. 200 ms | 70 |
| Amp Decay | `acid.amp_decay` | 50 .. 4000 ms | 900 |

### FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Cutoff | `acid.cutoff` | 80 .. 8000 Hz | 900 |
| Resonance | `acid.resonance` | 0 .. 1 | 0.72 |
| Env Amount | `acid.env_amount` | 0 .. 6 oct | 3.5 |
| Decay | `acid.decay` | 30 .. 2000 ms | 220 |
| Accent | `acid.accent` | 0 .. 1 | 0.7 |
| Key Track | `acid.key_track` | 0 .. 1 | 0.5 |

### SQUELCH & DRIVE

| Control | Key | Range | Default |
|---|---|---|---|
| Squelch | `acid.squelch` | off / on | off |
| Drive | `acid.drive` | 0 .. 1 | 0.85 |
| Squelch Start | `acid.squelch_start` | 2 .. 32 x | 12 |
| Squelch Time | `acid.squelch_time` | 5 .. 150 ms | 28 |
| Low Cut | `acid.low_cut` | 150 .. 1000 Hz | 250 |
| Comb Mix | `acid.comb_mix` | 0 .. 1 | 0.55 |
| Comb Feedback | `acid.comb_feedback` | 0 .. 0.97 | 0.82 |
| Disperse | `acid.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `acid.disperse_freq` | 200 .. 8000 Hz | 1250 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `acid.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `acid.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `acid.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `acid.menv_release` | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `acid.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `acid.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `acid.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `acid.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `acid.lfo1_fade` | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `acid.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `acid.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `acid.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `acid.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `acid.lfo2_fade` | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `acid.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `acid.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `acid.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `acid.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `acid.lfo3_fade` | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `acid.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `acid.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `acid.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `acid.lfo4_retrig` | off / on | off |

| | | | |
|---|---|---|---|
| LFO 4 Fade | acid.lfo4_fade | 0 .. 8 s | 0 |

## MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | acid.mx1_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | acid.mx1_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 1 Amount | acid.mx1_amount | -1 .. 1 | 0 |
| Mod 2 Source | acid.mx2_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | acid.mx2_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 2 Amount | acid.mx2_amount | -1 .. 1 | 0 |
| Mod 3 Source | acid.mx3_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | acid.mx3_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 3 Amount | acid.mx3_amount | -1 .. 1 | 0 |
| Mod 4 Source | acid.mx4_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | acid.mx4_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 4 Amount | acid.mx4_amount | -1 .. 1 | 0 |
| Mod 5 Source | acid.mx5_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | acid.mx5_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 5 Amount | acid.mx5_amount | -1 .. 1 | 0 |
| Mod 6 Source | acid.mx6_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | acid.mx6_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 6 Amount | acid.mx6_amount | -1 .. 1 | 0 |
| Mod 7 Source | acid.mx7_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | acid.mx7_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 7 Amount | acid.mx7_amount | -1 .. 1 | 0 |
| Mod 8 Source | acid.mx8_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | acid.mx8_dst | Off \| Pitch \| Cutoff \| Resonance \| Level \| Pan | Off |
| Mod 8 Amount | acid.mx8_amount | -1 .. 1 | 0 |

## DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | acid.delay_send | 0 .. 1 | 0.25 |
| Delay Feedback | acid.delay_feedback | 0 .. 0.9 | 0.5 |
| Delay Left | acid.delay_left | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/16 |
| Delay Right | acid.delay_right | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/4 |
| Delay High Pass | acid.delay_high_pass | 150 .. 2000 Hz | 400 |
| Delay Low Pass | acid.delay_low_pass | 800 .. 16000 Hz | 4500 |

## SPACE & OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | acid.level | -36 .. 6 dB | -2.3 |
| Duck | acid.duck | 0 .. 1 | 0.15 |
| Room Send | acid.room_send | 0 .. 1 | 0.1 |
| Plate Send | acid.plate_send | 0 .. 1 | 0.15 |
| Hall Send | acid.hall_send | 0 .. 1 | 0 |
| Hall Gate | acid.hall_gate | off / on | off |

## 9. Lead

A supersaw after Szabo's measurements of the JP-8000 -- the detune curve, the mix law and the random phase per note -- with a dynamic detune that follows the length of the note. The other three oscillator types are a virtual-analogue saw and pulse, two-operator FM and a wavetable.

Everything but the kick and the bass is high-passed at its own fundamental, which is what keeps eight voices of supersaw out of the bass register.

As in the acid, the Phrygian b2 is one of the lead's degrees, on beats 2 and 4 and off the beat; beats 1 and 3 take a tone of the tonic chord.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Lead Own Sound
Psy Pluck: Twilight Pollen (composer)
Preset
OSCILLATOR
Supersaw
Vocal
Oscillator
Table
0.55
0.75
0.00
0.60
0.30
Detune
Mix
Wave
Dynamic Detune
Position
0.50
Pulse Width
Drift
WAVETABLE MOTION
0.00
0.15
Position Env
Position LFO
Position Decay
LFO Period
FM
2.00
2.50
FM Ratio
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
0.28
Osc 2 Mix
Osc 2 Detune
-1 Oct
Osc 2 Interval
FILTER
State Variable
Low Pass
Filter Model
Filter Type
3.88 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
1897.21 Hz
Cutoff
0.50
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
0.55 ms
317.839 ms
0.05
158.92 ms
Attack
Decay
Sustain
Release
0.40
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
0.00
Mod Sustain
Mod Release
LFO 1
Sine
Free
1.00 Hz
LFO 1 Shape
LFO 1 Sync
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
Free
1.00 Hz
LFO 2 Shape
LFO 2 Sync
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
Free
1.00 Hz
LFO 3 Shape
LFO 3 Sync
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
Free
1.00 Hz
LFO 4 Shape
LFO 4 Sync
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
Velocity
Cutoff
0.34
Mod 1 Source
Mod 1 Target
Mod 1 Amount
Random
Cutoff
0.10
Mod 2 Source
Mod 2 Target
Mod 2 Amount
Off
Off
0.00
Mod 3 Source
Mod 3 Target
Mod 3 Amount
Off
Off
0.00
Mod 4 Source
Mod 4 Target
Mod 4 Amount
Off
Off
0.00
Mod 5 Source
Mod 5 Target
Mod 5 Amount
Off
Off
0.00
Mod 6 Source
Mod 6 Target
Mod 6 Amount
Off
Off
0.00
Mod 7 Source
Mod 7 Target
Mod 7 Amount
Off
Off
0.00
Mod 8 Source
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
8.00 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Phaser
Modulation
0.22
0.80
Mod Mix
Mod Depth
Mod Period
0.45
Mod Feedback
TRANCE GATE
Sixteenths
Trance Gate
Gate Pattern
0.85
0.50
Gate Depth
Gate Duty
Gate Attack
Gate Release
0.40
Gate Tone
DELAY
0.30
0.40
3/16
Delay Send
Delay Feedback
Delay Left
3/8
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
0.00
0.80
Distance
Width
Room Send
Plate Send
Hall Send
Hall Gate
-0.20
Pan
Slow Movement
OUTPUT
-4.00 dB
Level
0.20
Duck
PATTERN
LEAD bars 13 to 16
this part is silent here

The Lead tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Lead Own Sound | `mix.lead_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range |
|---|---|---|
| Oscillator | `lead.osc` | Supersaw \| VA \| FM \| Wavetable |
| Table | `lead.table` | Classic \| Vocal \| Glass \| PWM \| Sync \| Formant Saw \| WaveEdit Hienharm \| WaveEdit Junox_ho \| WaveEdit Euclidea \| WaveEdit Sohler49 \| WaveEdit Vocal_fo \| WaveEd |
| Position | `lead.position` | 0 .. 1 |
| Detune | `lead.detune` | 0 .. 1 |
| Mix | `lead.mix` | 0 .. 1 |
| Wave | `lead.wave` | 0 .. 1 |
| Dynamic Detune | `lead.dynamic_detune` | 0 .. 1 |
| Pulse Width | `lead.pulse_width` | 0.05 .. 0.95 |
| Drift | `lead.drift` | 0 .. 8 ct |

## WAVETABLE MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Position Env | `lead.pos_env` | -1 .. 1 | 0 |
| Position LFO | `lead.pos_lfo_depth` | 0 .. 0.5 | 0.15 |
| Position Decay | `lead.pos_decay` | 10 .. 8000 ms | 1500 |
| LFO Period | `lead.pos_lfo_beats` | 0.25 .. 64 beats | 16 |

## FM

| Control | Key | Range | Default |
|---|---|---|---|
| FM Ratio | `lead.fm_ratio` | 0.5 .. 8 | 2 |
| FM Index | `lead.fm_index` | 0 .. 10 | 2.5 |
| FM Decay | `lead.fm_decay` | 5 .. 2000 ms | 250 |

## SECOND OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Oscillator 2 | `lead.osc2` | Off \| Supersaw \| VA \| FM \| Wavetable | Off |
| Osc 2 Mix | `lead.osc2_mix` | 0 .. 1 | 0.28 |
| Osc 2 Interval | `lead.osc2_interval` | -2 Oct \| -1 Oct \| -5th \| Unison \| +5th \| +1 Oct | -1 Oct |
| Osc 2 Detune | `lead.osc2_detune` | -50 .. 50 ct | 7 |

## FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `lead.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `lead.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `lead.cutoff` | 200 .. 18000 Hz | 10000 |
| Resonance | `lead.resonance` | 0 .. 1 | 0.15 |
| Env Amount | `lead.env_amount` | 0 .. 6 oct | 2 |
| Filter Mode | `lead.filter_mode` | 0 .. 1 | 0 |
| Key Track | `lead.key_track` | 0 .. 1 | 0.5 |
| HP Floor | `lead.hp_floor` | 40 .. 400 Hz | 220 |
| HP Track | `lead.hp_track` | 0 .. 1 x f0 | 1 |
| Disperse | `lead.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `lead.disperse_freq` | 200 .. 8000 Hz | 1250 |

## FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `lead.filt_attack` | 0.1 .. 4000 ms | 0.1 |
| Filter Decay | `lead.filter_decay` | 5 .. 3000 ms | 400 |

| | | | |
|---|---|---|---|
| Filter Sustain | `lead.filt_sustain` | 0 .. 1 | 0 |
| Filter Release | `lead.filt_release` | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | `lead.amp_attack` | 0.3 .. 2000 ms | 4 |
| Decay | `lead.amp_decay` | 5 .. 4000 ms | 500 |
| Sustain | `lead.amp_sustain` | 0 .. 1 | 0.75 |
| Release | `lead.amp_release` | 5 .. 4000 ms | 180 |
| Velocity | `lead.vel_sens` | 0 .. 1 | 0.4 |
| Glide | `lead.glide` | 0 .. 400 ms | 45 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `lead.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `lead.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `lead.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `lead.menv_release` | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `lead.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `lead.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `lead.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `lead.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `lead.lfo1_fade` | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `lead.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `lead.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `lead.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `lead.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `lead.lfo2_fade` | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `lead.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `lead.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `lead.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `lead.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `lead.lfo3_fade` | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `lead.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `lead.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `lead.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `lead.lfo4_retrig` | off / on | off |
| LFO 4 Fade | `lead.lfo4_fade` | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | `lead.mx1_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | `lead.mx1_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 1 Amount | `lead.mx1_amount` | -1 .. 1 | 0 |
| Mod 2 Source | `lead.mx2_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | `lead.mx2_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |

| | | | |
|---|---|---|---|
| Mod 2 Amount | lead.mx2_amount | -1 .. 1 | 0 |
| Mod 3 Source | lead.mx3_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | lead.mx3_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 3 Amount | lead.mx3_amount | -1 .. 1 | 0 |
| Mod 4 Source | lead.mx4_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | lead.mx4_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 4 Amount | lead.mx4_amount | -1 .. 1 | 0 |
| Mod 5 Source | lead.mx5_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | lead.mx5_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 5 Amount | lead.mx5_amount | -1 .. 1 | 0 |
| Mod 6 Source | lead.mx6_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | lead.mx6_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 6 Amount | lead.mx6_amount | -1 .. 1 | 0 |
| Mod 7 Source | lead.mx7_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | lead.mx7_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 7 Amount | lead.mx7_amount | -1 .. 1 | 0 |
| Mod 8 Source | lead.mx8_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | lead.mx8_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 8 Amount | lead.mx8_amount | -1 .. 1 | 0 |

### VIBRATO & TREMOLO

| Control | Key | Range | Default |
|---|---|---|---|
| Voice LFO | lead.lfo_beats | 0.25 .. 64 beats | 8 |
| LFO to Cutoff | lead.lfo_cutoff | 0 .. 4 oct | 0.3 |
| LFO to Pitch | lead.lfo_pitch | 0 .. 50 ct | 0 |
| LFO to Level | lead.lfo_amp | 0 .. 1 | 0.06 |

### VOICE FX

| Control | Key | Range | Default |
|---|---|---|---|
| Modulation | lead.mod | Off \| Flanger \| Phaser \| Comb | Phaser |
| Mod Mix | lead.mod_mix | 0 .. 1 | 0.22 |
| Mod Depth | lead.mod_depth | 0 .. 1 | 0.8 |
| Mod Period | lead.mod_beats | 0.5 .. 64 beats | 16 |
| Mod Feedback | lead.mod_feedback | -0.9 .. 0.9 | 0.45 |

### TRANCE GATE

| Control | Key | Range | Default |
|---|---|---|---|
| Trance Gate | lead.gate | off / on | off |
| Gate Pattern | lead.gate_pattern | Sixteenths \| Eighths \| Rolling \| Gallop \| 3-3-2 \| Triplets | Sixteenths |
| Gate Depth | lead.gate_depth | 0 .. 1 | 0.85 |
| Gate Duty | lead.gate_duty | 0.05 .. 1 | 0.5 |
| Gate Attack | lead.gate_attack | 0.5 .. 60 ms | 3 |
| Gate Release | lead.gate_release | 0.5 .. 120 ms | 12 |
| Gate Tone | lead.gate_tone | 0 .. 1 | 0.4 |

### DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | lead.delay_send | 0 .. 1 | 0.3 |
| Delay Feedback | lead.delay_feedback | 0 .. 0.9 | 0.4 |
| Delay Left | lead.delay_left | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/16 |
| Delay Right | lead.delay_right | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/8 |
| Delay High Pass | lead.delay_high_pass | 150 .. 2000 Hz | 350 |
| Delay Low Pass | lead.delay_low_pass | 800 .. 16000 Hz | 6000 |

### SPACE & IMAGE

| Control | Key | Range | Default |
|---|---|---|---|
| Distance | lead.distance | 0 .. 1 | 0 |
| Width | lead.width | 0 .. 1 | 0.8 |

| | | | |
|---|---|---|---|
| Room Send | `lead.room_send` | 0 .. 1 | 0.1 |
| Plate Send | `lead.plate_send` | 0 .. 1 | 0 |
| Hall Send | `lead.hall_send` | 0 .. 1 | 0 |
| Hall Gate | `lead.hall_gate` | off / on | off |
| Pan | `lead.pan` | -1 .. 1 | -0.2 |
| Slow Movement | `lead.slow_mod` | 0 .. 1 | 0 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `lead.level` | -36 .. 6 dB | -4 |
| Duck | `lead.duck` | 0 .. 1 | 0.2 |

## 10. Counter

The counter-lead answers the lead. It plays in the gaps the lead leaves and in a register of its own, so the two never mask each other; Counter Mode on the Set tab (or the style, on Auto) decides whether it echoes, answers or holds against the lead. The engine is the same as the lead's, with its own knobs and its own presets.




The Counter tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Counter Own Sound | `mix.counter_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range |
|---|---|---|
| Oscillator | `counter.osc` | Supersaw \| VA \| FM \| Wavetable |
| Table | `counter.table` | Classic \| Vocal \| Glass \| PWM \| Sync \| Formant Saw \| WaveEdit Hienharm \| WaveEdit Junox_ho \| WaveEdit Euclidea \| WaveEdit Sohler49 \| WaveEdit Vocal_fo \| Wav |
| Position | `counter.position` | 0 .. 1 |
| Detune | `counter.detune` | 0 .. 1 |
| Mix | `counter.mix` | 0 .. 1 |
| Wave | `counter.wave` | 0 .. 1 |
| Dynamic Detune | `counter.dynamic_detune` | 0 .. 1 |
| Pulse Width | `counter.pulse_width` | 0.05 .. 0.95 |
| Drift | `counter.drift` | 0 .. 8 ct |

## WAVETABLE MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Position Env | `counter.pos_env` | -1 .. 1 | 0 |
| Position LFO | `counter.pos_lfo_depth` | 0 .. 0.5 | 0.25 |
| Position Decay | `counter.pos_decay` | 10 .. 8000 ms | 1500 |
| LFO Period | `counter.pos_lfo_beats` | 0.25 .. 64 beats | 4 |

## FM

| Control | Key | Range | Default |
|---|---|---|---|
| FM Ratio | `counter.fm_ratio` | 0.5 .. 8 | 2 |
| FM Index | `counter.fm_index` | 0 .. 10 | 2.5 |
| FM Decay | `counter.fm_decay` | 5 .. 2000 ms | 250 |

## SECOND OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Oscillator 2 | `counter.osc2` | Off \| Supersaw \| VA \| FM \| Wavetable | Off |
| Osc 2 Mix | `counter.osc2_mix` | 0 .. 1 | 0.3 |
| Osc 2 Interval | `counter.osc2_interval` | -2 Oct \| -1 Oct \| -5th \| Unison \| +5th \| +1 Oct | -1 Oct |
| Osc 2 Detune | `counter.osc2_detune` | -50 .. 50 ct | 6 |

## FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `counter.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `counter.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `counter.cutoff` | 200 .. 18000 Hz | 5000 |
| Resonance | `counter.resonance` | 0 .. 1 | 0.2 |
| Env Amount | `counter.env_amount` | 0 .. 6 oct | 3.2 |
| Filter Mode | `counter.filter_mode` | 0 .. 1 | 0 |
| Key Track | `counter.key_track` | 0 .. 1 | 0.5 |
| HP Floor | `counter.hp_floor` | 40 .. 400 Hz | 220 |
| HP Track | `counter.hp_track` | 0 .. 1 x f0 | 1 |
| Disperse | `counter.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `counter.disperse_freq` | 200 .. 8000 Hz | 1250 |

## FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `counter.filt_attack` | 0.1 .. 4000 ms | 0.1 |

| | | | |
|---|---|---|---|
| Filter Decay | counter.filter_decay | 5 .. 3000 ms | 60 |
| Filter Sustain | counter.filt_sustain | 0 .. 1 | 0 |
| Filter Release | counter.filt_release | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | counter.amp_attack | 0.3 .. 2000 ms | 1 |
| Decay | counter.amp_decay | 5 .. 4000 ms | 70 |
| Sustain | counter.amp_sustain | 0 .. 1 | 0 |
| Release | counter.amp_release | 5 .. 4000 ms | 35 |
| Velocity | counter.vel_sens | 0 .. 1 | 0.4 |
| Glide | counter.glide | 0 .. 400 ms | 0 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | counter.menv_attack | 0.1 .. 8000 ms | 10 |
| Mod Decay | counter.menv_decay | 5 .. 12000 ms | 800 |
| Mod Sustain | counter.menv_sustain | 0 .. 1 | 0 |
| Mod Release | counter.menv_release | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | counter.lfo1_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | counter.lfo1_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | counter.lfo1_rate | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | counter.lfo1_retrig | off / on | off |
| LFO 1 Fade | counter.lfo1_fade | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | counter.lfo2_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | counter.lfo2_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | counter.lfo2_rate | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | counter.lfo2_retrig | off / on | off |
| LFO 2 Fade | counter.lfo2_fade | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | counter.lfo3_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | counter.lfo3_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | counter.lfo3_rate | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | counter.lfo3_retrig | off / on | off |
| LFO 3 Fade | counter.lfo3_fade | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | counter.lfo4_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | counter.lfo4_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | counter.lfo4_rate | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | counter.lfo4_retrig | off / on | off |
| LFO 4 Fade | counter.lfo4_fade | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | counter.mx1_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | counter.mx1_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 1 Amount | counter.mx1_amount | -1 .. 1 | 0 |
| Mod 2 Source | counter.mx2_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |

| | | | |
|---|---|---|---|
| Mod 2 Target | counter.mx2_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 2 Amount | counter.mx2_amount | -1 .. 1 | 0 |
| Mod 3 Source | counter.mx3_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | counter.mx3_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 3 Amount | counter.mx3_amount | -1 .. 1 | 0 |
| Mod 4 Source | counter.mx4_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | counter.mx4_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 4 Amount | counter.mx4_amount | -1 .. 1 | 0 |
| Mod 5 Source | counter.mx5_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | counter.mx5_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 5 Amount | counter.mx5_amount | -1 .. 1 | 0 |
| Mod 6 Source | counter.mx6_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | counter.mx6_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 6 Amount | counter.mx6_amount | -1 .. 1 | 0 |
| Mod 7 Source | counter.mx7_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | counter.mx7_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 7 Amount | counter.mx7_amount | -1 .. 1 | 0 |
| Mod 8 Source | counter.mx8_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | counter.mx8_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 8 Amount | counter.mx8_amount | -1 .. 1 | 0 |

### VIBRATO & TREMOLO

| Control | Key | Range | Default |
|---|---|---|---|
| Voice LFO | counter.lfo_beats | 0.25 .. 64 beats | 4 |
| LFO to Cutoff | counter.lfo_cutoff | 0 .. 4 oct | 0.35 |
| LFO to Pitch | counter.lfo_pitch | 0 .. 50 ct | 0 |
| LFO to Level | counter.lfo_amp | 0 .. 1 | 0 |

### VOICE FX

| Control | Key | Range | Default |
|---|---|---|---|
| Modulation | counter.mod | Off \| Flanger \| Phaser \| Comb | Flanger |
| Mod Mix | counter.mod_mix | 0 .. 1 | 0.28 |
| Mod Depth | counter.mod_depth | 0 .. 1 | 0.75 |
| Mod Period | counter.mod_beats | 0.5 .. 64 beats | 4 |
| Mod Feedback | counter.mod_feedback | -0.9 .. 0.9 | 0.55 |

### TRANCE GATE

| Control | Key | Range | Default |
|---|---|---|---|
| Trance Gate | counter.gate | off / on | off |
| Gate Pattern | counter.gate_pattern | Sixteenths \| Eighths \| Rolling \| Gallop \| 3-3-2 \| Triplets | Sixteenths |
| Gate Depth | counter.gate_depth | 0 .. 1 | 0.85 |
| Gate Duty | counter.gate_duty | 0.05 .. 1 | 0.5 |
| Gate Attack | counter.gate_attack | 0.5 .. 60 ms | 3 |
| Gate Release | counter.gate_release | 0.5 .. 120 ms | 12 |
| Gate Tone | counter.gate_tone | 0 .. 1 | 0.4 |

### DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | counter.delay_send | 0 .. 1 | 0.4 |
| Delay Feedback | counter.delay_feedback | 0 .. 0.9 | 0.4 |
| Delay Left | counter.delay_left | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/16 |
| Delay Right | counter.delay_right | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/8 |
| Delay High Pass | counter.delay_high_pass | 150 .. 2000 Hz | 350 |
| Delay Low Pass | counter.delay_low_pass | 800 .. 16000 Hz | 6000 |

### SPACE & IMAGE

| Control | Key | Range | Default |
|---|---|---|---|
| Distance | counter.distance | 0 .. 1 | 0.5 |

| | | | |
|---|---|---|---|
| Width | `counter.width` | 0 .. 1 | 0.7 |
| Room Send | `counter.room_send` | 0 .. 1 | 0 |
| Plate Send | `counter.plate_send` | 0 .. 1 | 0.2 |
| Hall Send | `counter.hall_send` | 0 .. 1 | 0 |
| Hall Gate | `counter.hall_gate` | off / on | off |
| Pan | `counter.pan` | -1 .. 1 | 0.2 |
| Slow Movement | `counter.slow_mod` | 0 .. 1 | 0 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `counter.level` | -36 .. 6 dB | -3 |
| Duck | `counter.duck` | 0 .. 1 | 0.2 |

## 11. Arp

The same engine as the lead, with its own knobs. The arpeggio runs on chord tones, and when its pitch range would overlap the lead's by more than two semitones it is octaved away: two voices playing in the same two octaves mask each other, and the composer checks that before it writes the notes.

The counter never strikes together with the lead a semitone or a minor ninth apart on beats 1 and 3.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Arp Own Sound
Rolling Sequence: Deep Turbine (composer)
Preset
OSCILLATOR
VA
Oscillator
Vocal
Table
0.30
Detune
0.60
Mix
0.39
Wave
0.60
Dynamic Detune
0.50
Pulse Width
0.30
Position
WAVETABLE MOTION
0.00
Position Env
0.15
Position LFO
Position Decay
LFO Period
FM
2.00
FM Ratio
2.50
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
-1 Oct
Osc 2 Interval
0.26
Osc 2 Mix
Osc 2 Detune
FILTER
Prophet (SSM2040)
Filter Model
Low Pass
Filter Type
3.91 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
700 Hz
Cutoff
0.29
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
0.67 ms
Attack
100.68 ms
Decay
0.03
Sustain
50.3 ms
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
LFO 1 Shape
Free
LFO 1 Sync
1.00 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
LFO 2 Shape
Free
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
LFO 3 Shape
Free
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
LFO 4 Shape
Free
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
Off
Mod 1 Source
Off
Mod 1 Target
Mod 1 Amount
Off
Mod 2 Source
Off
Mod 2 Target
Mod 2 Amount
Off
Mod 3 Source
Off
Mod 3 Target
Mod 3 Amount
Off
Mod 4 Source
Off
Mod 4 Target
Mod 4 Amount
Off
Mod 5 Source
Off
Mod 5 Target
Mod 5 Amount
Off
Mod 6 Source
Off
Mod 6 Target
Mod 6 Amount
Off
Mod 7 Source
Off
Mod 7 Target
Mod 7 Amount
Off
Mod 8 Source
Off
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
8.00 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Comb
Modulation
-0.20
Mod Mix
0.00
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
Gate Depth
0.50
Gate Duty
Gate Attack
Gate Release
0.40
Gate Tone
DELAY
0.35
Delay Send
0.40
Delay Feedback
3/16
Delay Left
1/8
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
0.50
Distance
0.60
Width
Room Send
Plate Send
Hall Send
Hall Gate
Pan
Slow Movement
OUTPUT
-12.4 dB
Level
-0.20
Duck
PATTERN
ARP bars 17 to 20
this part is silent here

The Arp tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Arp Own Sound | `mix.arp_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range |
|---|---|---|
| Oscillator | `arp.osc` | Supersaw \| VA \| FM \| Wavetable |
| Table | `arp.table` | Classic \| Vocal \| Glass \| PWM \| Sync \| Formant Saw \| WaveEdit Hienharm \| WaveEdit Junox_ho \| WaveEdit Euclidea \| WaveEdit Sohler49 \| WaveEdit Vocal_fo \| WaveEdit |
| Position | `arp.position` | 0 .. 1 |
| Detune | `arp.detune` | 0 .. 1 |
| Mix | `arp.mix` | 0 .. 1 |
| Wave | `arp.wave` | 0 .. 1 |
| Dynamic Detune | `arp.dynamic_detune` | 0 .. 1 |
| Pulse Width | `arp.pulse_width` | 0.05 .. 0.95 |
| Drift | `arp.drift` | 0 .. 8 ct |

## WAVETABLE MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Position Env | `arp.pos_env` | -1 .. 1 | 0 |
| Position LFO | `arp.pos_lfo_depth` | 0 .. 0.5 | 0.15 |
| Position Decay | `arp.pos_decay` | 10 .. 8000 ms | 1500 |
| LFO Period | `arp.pos_lfo_beats` | 0.25 .. 64 beats | 16 |

## FM

| Control | Key | Range | Default |
|---|---|---|---|
| FM Ratio | `arp.fm_ratio` | 0.5 .. 8 | 2 |
| FM Index | `arp.fm_index` | 0 .. 10 | 2.5 |
| FM Decay | `arp.fm_decay` | 5 .. 2000 ms | 250 |

## SECOND OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Oscillator 2 | `arp.osc2` | Off \| Supersaw \| VA \| FM \| Wavetable | Off |
| Osc 2 Mix | `arp.osc2_mix` | 0 .. 1 | 0.26 |
| Osc 2 Interval | `arp.osc2_interval` | -2 Oct \| -1 Oct \| -5th \| Unison \| +5th \| +1 Oct | -1 Oct |
| Osc 2 Detune | `arp.osc2_detune` | -50 .. 50 ct | 4 |

## FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `arp.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `arp.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `arp.cutoff` | 200 .. 18000 Hz | 3500 |
| Resonance | `arp.resonance` | 0 .. 1 | 0.3 |
| Env Amount | `arp.env_amount` | 0 .. 6 oct | 2.8 |
| Filter Mode | `arp.filter_mode` | 0 .. 1 | 0 |
| Key Track | `arp.key_track` | 0 .. 1 | 0.5 |
| HP Floor | `arp.hp_floor` | 40 .. 400 Hz | 220 |
| HP Track | `arp.hp_track` | 0 .. 1 x f0 | 1 |
| Disperse | `arp.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `arp.disperse_freq` | 200 .. 8000 Hz | 1250 |

## FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `arp.filt_attack` | 0.1 .. 4000 ms | 0.1 |
| Filter Decay | `arp.filter_decay` | 5 .. 3000 ms | 140 |
| Filter Sustain | `arp.filt_sustain` | 0 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Filter Release | `arp.filt_release` | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | `arp.amp_attack` | 0.3 .. 2000 ms | 0.8 |
| Decay | `arp.amp_decay` | 5 .. 4000 ms | 220 |
| Sustain | `arp.amp_sustain` | 0 .. 1 | 0 |
| Release | `arp.amp_release` | 5 .. 4000 ms | 12 |
| Velocity | `arp.vel_sens` | 0 .. 1 | 0.4 |
| Glide | `arp.glide` | 0 .. 400 ms | 0 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `arp.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `arp.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `arp.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `arp.menv_release` | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `arp.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `arp.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `arp.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `arp.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `arp.lfo1_fade` | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `arp.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `arp.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `arp.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `arp.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `arp.lfo2_fade` | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `arp.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `arp.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `arp.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `arp.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `arp.lfo3_fade` | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `arp.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `arp.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `arp.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `arp.lfo4_retrig` | off / on | off |
| LFO 4 Fade | `arp.lfo4_fade` | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | `arp.mx1_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | `arp.mx1_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 1 Amount | `arp.mx1_amount` | -1 .. 1 | 0 |
| Mod 2 Source | `arp.mx2_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | `arp.mx2_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 2 Amount | `arp.mx2_amount` | -1 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Mod 3 Source | arp.mx3_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | arp.mx3_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 3 Amount | arp.mx3_amount | -1 .. 1 | 0 |
| Mod 4 Source | arp.mx4_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | arp.mx4_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 4 Amount | arp.mx4_amount | -1 .. 1 | 0 |
| Mod 5 Source | arp.mx5_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | arp.mx5_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 5 Amount | arp.mx5_amount | -1 .. 1 | 0 |
| Mod 6 Source | arp.mx6_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | arp.mx6_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 6 Amount | arp.mx6_amount | -1 .. 1 | 0 |
| Mod 7 Source | arp.mx7_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | arp.mx7_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 7 Amount | arp.mx7_amount | -1 .. 1 | 0 |
| Mod 8 Source | arp.mx8_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | arp.mx8_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 8 Amount | arp.mx8_amount | -1 .. 1 | 0 |

### VIBRATO & TREMOLO

| Control | Key | Range | Default |
|---|---|---|---|
| Voice LFO | arp.lfo_beats | 0.25 .. 64 beats | 8 |
| LFO to Cutoff | arp.lfo_cutoff | 0 .. 4 oct | 0.28 |
| LFO to Pitch | arp.lfo_pitch | 0 .. 50 ct | 0 |
| LFO to Level | arp.lfo_amp | 0 .. 1 | 0.08 |

### VOICE FX

| Control | Key | Range | Default |
|---|---|---|---|
| Modulation | arp.mod | Off \| Flanger \| Phaser \| Comb | Comb |
| Mod Mix | arp.mod_mix | 0 .. 1 | 0.2 |
| Mod Depth | arp.mod_depth | 0 .. 1 | 0 |
| Mod Period | arp.mod_beats | 0.5 .. 64 beats | 16 |
| Mod Feedback | arp.mod_feedback | -0.9 .. 0.9 | 0.6 |

### TRANCE GATE

| Control | Key | Range | Default |
|---|---|---|---|
| Trance Gate | arp.gate | off / on | off |
| Gate Pattern | arp.gate_pattern | Sixteenths \| Eighths \| Rolling \| Gallop \| 3-3-2 \| Triplets | Sixteenths |
| Gate Depth | arp.gate_depth | 0 .. 1 | 0.85 |
| Gate Duty | arp.gate_duty | 0.05 .. 1 | 0.5 |
| Gate Attack | arp.gate_attack | 0.5 .. 60 ms | 3 |
| Gate Release | arp.gate_release | 0.5 .. 120 ms | 12 |
| Gate Tone | arp.gate_tone | 0 .. 1 | 0.4 |

### DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | arp.delay_send | 0 .. 1 | 0.35 |
| Delay Feedback | arp.delay_feedback | 0 .. 0.9 | 0.4 |
| Delay Left | arp.delay_left | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/16 |
| Delay Right | arp.delay_right | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/8 |
| Delay High Pass | arp.delay_high_pass | 150 .. 2000 Hz | 350 |
| Delay Low Pass | arp.delay_low_pass | 800 .. 16000 Hz | 6000 |

### SPACE & IMAGE

| Control | Key | Range | Default |
|---|---|---|---|
| Distance | arp.distance | 0 .. 1 | 0.5 |
| Width | arp.width | 0 .. 1 | 0.6 |
| Room Send | arp.room_send | 0 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Plate Send | `arp.plate_send` | 0 .. 1 | 0.2 |
| Hall Send | `arp.hall_send` | 0 .. 1 | 0 |
| Hall Gate | `arp.hall_gate` | off / on | off |
| Pan | `arp.pan` | -1 .. 1 | 0 |
| Slow Movement | `arp.slow_mod` | 0 .. 1 | 0 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `arp.level` | -36 .. 6 dB | -5 |
| Duck | `arp.duck` | 0 .. 1 | 0.2 |

## 12. Stab

Short chord hits on the off-beats and the phrase ends, the punctuation of a groove. They are voiced from the track's harmony and kept out of the lead's register.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Stab Own Sound
Supersaw Stab: Buried Torch (composer)
Preset
OSCILLATOR
Supersaw
Oscillator
Vocal
Table
0.74
Detune
0.83
Mix
0.00
Wave
Dynamic Detune
Pulse Width
Drift
0.30
Position
WAVETABLE MOTION
0.00
Position Env
0.15
Position LFO
Position Decay
LFO Period
FM
2.00
FM Ratio
2.50
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
-1 Oct
Osc 2 Interval
0.26
Osc 2 Mix
Osc 2 Detune
FILTER
Prophet (SSM2040)
Filter Model
Low Pass
Filter Type
1.93 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
2500 Hz
Cutoff
0.13
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
2.84 ms
Attack
395.284 ms
Decay
0.83
Sustain
332.574 ms
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
LFO 1 Shape
Free
LFO 1 Sync
0.13 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
LFO 2 Shape
Free
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
LFO 3 Shape
Free
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
LFO 4 Shape
Free
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
LFO 1
Mod 1 Source
Cutoff
Mod 1 Target
Mod 1 Amount
Velocity
Mod 2 Source
Cutoff
Mod 2 Target
Mod 2 Amount
Off
Mod 3 Source
Off
Mod 3 Target
Mod 3 Amount
Off
Mod 4 Source
Off
Mod 4 Target
Mod 4 Amount
Off
Mod 5 Source
Off
Mod 5 Target
Mod 5 Amount
Off
Mod 6 Source
Off
Mod 6 Target
Mod 6 Amount
Off
Mod 7 Source
Off
Mod 7 Target
Mod 7 Amount
Off
Mod 8 Source
Off
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
8.00 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
LFO to Level
VOICE FX
Off
Modulation
0.00
Mod Mix
0.70
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
Gate Depth
Gate Duty
Gate Attack
Gate Release
Gate Tone
DELAY
0.45
Delay Send
0.45
Delay Feedback
3/16
Delay Left
1/4
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
0.50
Distance
0.90
Width
Room Send
Plate Send
Hall Send
Hall Gate
Pan
Slow Movement
OUTPUT
-7.00 dB
Level
0.20
Duck
PATTERN
STAB bars 17 to 20
this part is silent here

The Stab tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Stab Own Sound | `mix.stab_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range |
|---|---|---|
| Oscillator | `stab.osc` | Supersaw \| VA \| FM \| Wavetable |
| Table | `stab.table` | Classic \| Vocal \| Glass \| PWM \| Sync \| Formant Saw \| WaveEdit Hienharm \| WaveEdit Junox_ho \| WaveEdit Euclidea \| WaveEdit Sohler49 \| WaveEdit Vocal_fo \| WaveEd |
| Position | `stab.position` | 0 .. 1 |
| Detune | `stab.detune` | 0 .. 1 |
| Mix | `stab.mix` | 0 .. 1 |
| Wave | `stab.wave` | 0 .. 1 |
| Dynamic Detune | `stab.dynamic_detune` | 0 .. 1 |
| Pulse Width | `stab.pulse_width` | 0.05 .. 0.95 |
| Drift | `stab.drift` | 0 .. 8 ct |

## WAVETABLE MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Position Env | `stab.pos_env` | -1 .. 1 | 0 |
| Position LFO | `stab.pos_lfo_depth` | 0 .. 0.5 | 0.15 |
| Position Decay | `stab.pos_decay` | 10 .. 8000 ms | 1500 |
| LFO Period | `stab.pos_lfo_beats` | 0.25 .. 64 beats | 16 |

## FM

| Control | Key | Range | Default |
|---|---|---|---|
| FM Ratio | `stab.fm_ratio` | 0.5 .. 8 | 2 |
| FM Index | `stab.fm_index` | 0 .. 10 | 2.5 |
| FM Decay | `stab.fm_decay` | 5 .. 2000 ms | 250 |

## SECOND OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Oscillator 2 | `stab.osc2` | Off \| Supersaw \| VA \| FM \| Wavetable | Off |
| Osc 2 Mix | `stab.osc2_mix` | 0 .. 1 | 0.26 |
| Osc 2 Interval | `stab.osc2_interval` | -2 Oct \| -1 Oct \| -5th \| Unison \| +5th \| +1 Oct | -1 Oct |
| Osc 2 Detune | `stab.osc2_detune` | -50 .. 50 ct | 5 |

## FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `stab.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `stab.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `stab.cutoff` | 200 .. 18000 Hz | 1800 |
| Resonance | `stab.resonance` | 0 .. 1 | 0.35 |
| Env Amount | `stab.env_amount` | 0 .. 6 oct | 3.5 |
| Filter Mode | `stab.filter_mode` | 0 .. 1 | 0 |
| Key Track | `stab.key_track` | 0 .. 1 | 0.5 |
| HP Floor | `stab.hp_floor` | 40 .. 400 Hz | 220 |
| HP Track | `stab.hp_track` | 0 .. 1 x f0 | 1 |
| Disperse | `stab.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `stab.disperse_freq` | 200 .. 8000 Hz | 1250 |

## FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `stab.filt_attack` | 0.1 .. 4000 ms | 0.1 |
| Filter Decay | `stab.filter_decay` | 5 .. 3000 ms | 90 |

| | | | |
|---|---|---|---|
| Filter Sustain | `stab.filt_sustain` | 0 .. 1 | 0 |
| Filter Release | `stab.filt_release` | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | `stab.amp_attack` | 0.3 .. 2000 ms | 0.5 |
| Decay | `stab.amp_decay` | 5 .. 4000 ms | 180 |
| Sustain | `stab.amp_sustain` | 0 .. 1 | 0 |
| Release | `stab.amp_release` | 5 .. 4000 ms | 60 |
| Velocity | `stab.vel_sens` | 0 .. 1 | 0.4 |
| Glide | `stab.glide` | 0 .. 400 ms | 0 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `stab.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `stab.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `stab.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `stab.menv_release` | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `stab.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `stab.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `stab.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `stab.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `stab.lfo1_fade` | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `stab.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `stab.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `stab.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `stab.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `stab.lfo2_fade` | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `stab.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `stab.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `stab.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `stab.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `stab.lfo3_fade` | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `stab.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `stab.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `stab.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `stab.lfo4_retrig` | off / on | off |
| LFO 4 Fade | `stab.lfo4_fade` | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | `stab.mx1_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | `stab.mx1_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 1 Amount | `stab.mx1_amount` | -1 .. 1 | 0 |
| Mod 2 Source | `stab.mx2_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | `stab.mx2_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |

| | | | |
|---|---|---|---|
| Mod 2 Amount | stab.mx2_amount | -1 .. 1 | 0 |
| Mod 3 Source | stab.mx3_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | stab.mx3_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 3 Amount | stab.mx3_amount | -1 .. 1 | 0 |
| Mod 4 Source | stab.mx4_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | stab.mx4_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 4 Amount | stab.mx4_amount | -1 .. 1 | 0 |
| Mod 5 Source | stab.mx5_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | stab.mx5_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 5 Amount | stab.mx5_amount | -1 .. 1 | 0 |
| Mod 6 Source | stab.mx6_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | stab.mx6_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 6 Amount | stab.mx6_amount | -1 .. 1 | 0 |
| Mod 7 Source | stab.mx7_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | stab.mx7_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 7 Amount | stab.mx7_amount | -1 .. 1 | 0 |
| Mod 8 Source | stab.mx8_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | stab.mx8_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 8 Amount | stab.mx8_amount | -1 .. 1 | 0 |

### VIBRATO & TREMOLO

| Control | Key | Range | Default |
|---|---|---|---|
| Voice LFO | stab.lfo_beats | 0.25 .. 64 beats | 8 |
| LFO to Cutoff | stab.lfo_cutoff | 0 .. 4 oct | 0.22 |
| LFO to Pitch | stab.lfo_pitch | 0 .. 50 ct | 0 |
| LFO to Level | stab.lfo_amp | 0 .. 1 | 0 |

### VOICE FX

| Control | Key | Range | Default |
|---|---|---|---|
| Modulation | stab.mod | Off \| Flanger \| Phaser \| Comb | Off |
| Mod Mix | stab.mod_mix | 0 .. 1 | 0 |
| Mod Depth | stab.mod_depth | 0 .. 1 | 0.7 |
| Mod Period | stab.mod_beats | 0.5 .. 64 beats | 16 |
| Mod Feedback | stab.mod_feedback | -0.9 .. 0.9 | 0.5 |

### TRANCE GATE

| Control | Key | Range | Default |
|---|---|---|---|
| Trance Gate | stab.gate | off / on | off |
| Gate Pattern | stab.gate_pattern | Sixteenths \| Eighths \| Rolling \| Gallop \| 3-3-2 \| Triplets | Sixteenths |
| Gate Depth | stab.gate_depth | 0 .. 1 | 0.85 |
| Gate Duty | stab.gate_duty | 0.05 .. 1 | 0.5 |
| Gate Attack | stab.gate_attack | 0.5 .. 60 ms | 3 |
| Gate Release | stab.gate_release | 0.5 .. 120 ms | 12 |
| Gate Tone | stab.gate_tone | 0 .. 1 | 0.4 |

### DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | stab.delay_send | 0 .. 1 | 0.45 |
| Delay Feedback | stab.delay_feedback | 0 .. 0.9 | 0.45 |
| Delay Left | stab.delay_left | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/16 |
| Delay Right | stab.delay_right | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/4 |
| Delay High Pass | stab.delay_high_pass | 150 .. 2000 Hz | 350 |
| Delay Low Pass | stab.delay_low_pass | 800 .. 16000 Hz | 6000 |

### SPACE & IMAGE

| Control | Key | Range | Default |
|---|---|---|---|
| Distance | stab.distance | 0 .. 1 | 0.5 |
| Width | stab.width | 0 .. 1 | 0.9 |

| | | | |
|---|---|---|---|
| Room Send | `stab.room_send` | 0 .. 1 | 0 |
| Plate Send | `stab.plate_send` | 0 .. 1 | 0.15 |
| Hall Send | `stab.hall_send` | 0 .. 1 | 0 |
| Hall Gate | `stab.hall_gate` | off / on | off |
| Pan | `stab.pan` | -1 .. 1 | 0 |
| Slow Movement | `stab.slow_mod` | 0 .. 1 | 0 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `stab.level` | -36 .. 6 dB | -7 |
| Duck | `stab.duck` | 0 .. 1 | 0.2 |

## 13. Pad

Four-voice voicings with the smallest voice movement between chords, through the wavetable engine. The pads carry the blocks that have no lead and no acid, and the trance gate gives them their movement.

Every pitched voice plays its FM ratio to the nearest half (1, 1.5, 2, ...), so an FM pad or lead is always harmonic: a ratio in between makes the bell and struck-metal spectra that sounded out of tune under a chord. A second oscillator stays within a few cents of its interval, and the pad's partner plays in unison or an octave up, never a fifth, which would stack a second chord on the first.

A pad never holds a minor ninth -- no note a semitone above the tonic that bass and drone play, none a semitone above its own root -- nor the leading tone under the tonic, nor a tritone over its root. Held for bars, those intervals sound wrong rather than dark; the genre makes its dissonance with the sound (clusters, noise, drones), and the Phrygian b2 belongs to the lead and the acid. Where the progression swings to the bII, the pad stays on the tonic and adds the minor sixth instead (E - B - E - C over E). The major seventh is a rare colour, on bVI and bIII only.


PHOSPHENE
Progressive
F#
Harmonic Minor
Set
60 min
Compose set
seed 112484790 145.0 BPM own clock playing
track 1, bar 16 (bar 16 of the set)
ARRANGE 24 tracks 5456 bars planned set 60 min
Set
Arrange
Low End
Percussion
Acid
Synths
Effects
Mixer
Perform
Export
Gallery
Lead
Counter
Arp
Stab
Pad
Drone
SOUND
Pad Own Sound
Supersaw Pad: Twilight Stupa (composer)
Preset
OSCILLATOR
Supersaw
Oscillator
Vocal
Table
0.85
Detune
0.69
Mix
0.00
Wave
Dynamic Detune
Pulse Width
Drift
0.30
Position
WAVETABLE MOTION
0.30
Position Env
0.15
Position LFO
Position Decay
LFO Period
FM
2.00
FM Ratio
2.50
FM Index
FM Decay
SECOND OSCILLATOR
Off
Oscillator 2
-1 Oct
Osc 2 Interval
0.38
Osc 2 Mix
Osc 2 Detune
FILTER
Juno (IR3109)
Filter Model
Low Pass
Filter Type
2.50 oct
Env Amount
Filter Mode
Key Track
HP Floor
HP Track
Disperse
Disperse Freq
4248.91 Hz
Cutoff
0.24
Resonance
FILTER ENVELOPE
Filter Attack
Filter Decay
Filter Sustain
Filter Release
AMPLIFIER
2.65 ms
Attack
776.995 ms
Decay
0.74
Sustain
500 ms
Release
Velocity
Glide
MOD ENVELOPE
Mod Attack
Mod Decay
Mod Sustain
Mod Release
LFO 1
Sine
LFO 1 Shape
Free
LFO 1 Sync
1.00 Hz
LFO 1 Rate
LFO 1 Retrig
LFO 1 Fade
LFO 2
Sine
LFO 2 Shape
Free
LFO 2 Sync
1.00 Hz
LFO 2 Rate
LFO 2 Retrig
LFO 2 Fade
LFO 3
Sine
LFO 3 Shape
Free
LFO 3 Sync
1.00 Hz
LFO 3 Rate
LFO 3 Retrig
LFO 3 Fade
LFO 4
Sine
LFO 4 Shape
Free
LFO 4 Sync
1.00 Hz
LFO 4 Rate
LFO 4 Retrig
LFO 4 Fade
MOD MATRIX
Off
Mod 1 Source
Off
Mod 1 Target
Mod 1 Amount
Off
Mod 2 Source
Off
Mod 2 Target
Mod 2 Amount
Off
Mod 3 Source
Off
Mod 3 Target
Mod 3 Amount
Off
Mod 4 Source
Off
Mod 4 Target
Mod 4 Amount
Off
Mod 5 Source
Off
Mod 5 Target
Mod 5 Amount
Off
Mod 6 Source
Off
Mod 6 Target
Mod 6 Amount
Off
Mod 7 Source
Off
Mod 7 Target
Mod 7 Amount
Off
Mod 8 Source
Off
Mod 8 Target
Mod 8 Amount
VIBRATO & TREMOLO
16.0 beats
Voice LFO
LFO to Cutoff
LFO to Pitch
0.10
LFO to Level
VOICE FX
Off
Modulation
0.00
Mod Mix
0.70
Mod Depth
Mod Period
Mod Feedback
TRANCE GATE
Trance Gate
Sixteenths
Gate Pattern
0.85
Gate Depth
0.50
Gate Duty
Gate Attack
Gate Release
0.40
Gate Tone
DELAY
0.00
Delay Send
0.40
Delay Feedback
3/16
Delay Left
1/4
Delay Right
Delay High Pass
Delay Low Pass
SPACE & IMAGE
1.00
Distance
1.00
Width
Room Send
Plate Send
0.34
Hall Send
Hall Gate
Pan
0.50
Slow Movement
OUTPUT
-16.0 dB
Level
0.50
Duck
PATTERN
PAD bars 17 to 20

The Pad tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Pad Own Sound | `mix.pad_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range |
|---|---|---|
| Oscillator | `pad.osc` | Supersaw \| VA \| FM \| Wavetable |
| Table | `pad.table` | Classic \| Vocal \| Glass \| PWM \| Sync \| Formant Saw \| WaveEdit Hienharm \| WaveEdit Junox_ho \| WaveEdit Euclidea \| WaveEdit Sohler49 \| WaveEdit Vocal_fo \| WaveEdit |
| Position | `pad.position` | 0 .. 1 |
| Detune | `pad.detune` | 0 .. 1 |
| Mix | `pad.mix` | 0 .. 1 |
| Wave | `pad.wave` | 0 .. 1 |
| Dynamic Detune | `pad.dynamic_detune` | 0 .. 1 |
| Pulse Width | `pad.pulse_width` | 0.05 .. 0.95 |
| Drift | `pad.drift` | 0 .. 8 ct |

## WAVETABLE MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Position Env | `pad.pos_env` | -1 .. 1 | 0.3 |
| Position LFO | `pad.pos_lfo_depth` | 0 .. 0.5 | 0.15 |
| Position Decay | `pad.pos_decay` | 10 .. 8000 ms | 3000 |
| LFO Period | `pad.pos_lfo_beats` | 0.25 .. 64 beats | 16 |

## FM

| Control | Key | Range | Default |
|---|---|---|---|
| FM Ratio | `pad.fm_ratio` | 0.5 .. 8 | 2 |
| FM Index | `pad.fm_index` | 0 .. 10 | 2.5 |
| FM Decay | `pad.fm_decay` | 5 .. 2000 ms | 250 |

## SECOND OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Oscillator 2 | `pad.osc2` | Off \| Supersaw \| VA \| FM \| Wavetable | Off |
| Osc 2 Mix | `pad.osc2_mix` | 0 .. 1 | 0.38 |
| Osc 2 Interval | `pad.osc2_interval` | -2 Oct \| -1 Oct \| -5th \| Unison \| +5th \| +1 Oct | -1 Oct |
| Osc 2 Detune | `pad.osc2_detune` | -50 .. 50 ct | 9 |

## FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `pad.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `pad.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `pad.cutoff` | 200 .. 18000 Hz | 5000 |
| Resonance | `pad.resonance` | 0 .. 1 | 0.1 |
| Env Amount | `pad.env_amount` | 0 .. 6 oct | 0 |
| Filter Mode | `pad.filter_mode` | 0 .. 1 | 0 |
| Key Track | `pad.key_track` | 0 .. 1 | 0.5 |
| HP Floor | `pad.hp_floor` | 40 .. 400 Hz | 140 |
| HP Track | `pad.hp_track` | 0 .. 1 x f0 | 1 |
| Disperse | `pad.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `pad.disperse_freq` | 200 .. 8000 Hz | 1250 |

## FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `pad.filt_attack` | 0.1 .. 4000 ms | 0.1 |
| Filter Decay | `pad.filter_decay` | 5 .. 3000 ms | 400 |
| Filter Sustain | `pad.filt_sustain` | 0 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Filter Release | `pad.filt_release` | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | `pad.amp_attack` | 0.3 .. 2000 ms | 700 |
| Decay | `pad.amp_decay` | 5 .. 4000 ms | 2000 |
| Sustain | `pad.amp_sustain` | 0 .. 1 | 1 |
| Release | `pad.amp_release` | 5 .. 4000 ms | 1800 |
| Velocity | `pad.vel_sens` | 0 .. 1 | 0.4 |
| Glide | `pad.glide` | 0 .. 400 ms | 0 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `pad.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `pad.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `pad.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `pad.menv_release` | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `pad.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `pad.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `pad.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `pad.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `pad.lfo1_fade` | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `pad.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `pad.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `pad.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `pad.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `pad.lfo2_fade` | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `pad.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `pad.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `pad.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `pad.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `pad.lfo3_fade` | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `pad.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `pad.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `pad.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `pad.lfo4_retrig` | off / on | off |
| LFO 4 Fade | `pad.lfo4_fade` | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | `pad.mx1_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | `pad.mx1_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 1 Amount | `pad.mx1_amount` | -1 .. 1 | 0 |
| Mod 2 Source | `pad.mx2_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | `pad.mx2_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 2 Amount | `pad.mx2_amount` | -1 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Mod 3 Source | `pad.mx3_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | `pad.mx3_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 3 Amount | `pad.mx3_amount` | -1 .. 1 | 0 |
| Mod 4 Source | `pad.mx4_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | `pad.mx4_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 4 Amount | `pad.mx4_amount` | -1 .. 1 | 0 |
| Mod 5 Source | `pad.mx5_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | `pad.mx5_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 5 Amount | `pad.mx5_amount` | -1 .. 1 | 0 |
| Mod 6 Source | `pad.mx6_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | `pad.mx6_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 6 Amount | `pad.mx6_amount` | -1 .. 1 | 0 |
| Mod 7 Source | `pad.mx7_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | `pad.mx7_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 7 Amount | `pad.mx7_amount` | -1 .. 1 | 0 |
| Mod 8 Source | `pad.mx8_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | `pad.mx8_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 8 Amount | `pad.mx8_amount` | -1 .. 1 | 0 |

## VIBRATO & TREMOLO

| Control | Key | Range | Default |
|---|---|---|---|
| Voice LFO | `pad.lfo_beats` | 0.25 .. 64 beats | 16 |
| LFO to Cutoff | `pad.lfo_cutoff` | 0 .. 4 oct | 0.5 |
| LFO to Pitch | `pad.lfo_pitch` | 0 .. 50 ct | 0 |
| LFO to Level | `pad.lfo_amp` | 0 .. 1 | 0.1 |

## VOICE FX

| Control | Key | Range | Default |
|---|---|---|---|
| Modulation | `pad.mod` | Off \| Flanger \| Phaser \| Comb | Off |
| Mod Mix | `pad.mod_mix` | 0 .. 1 | 0 |
| Mod Depth | `pad.mod_depth` | 0 .. 1 | 0.7 |
| Mod Period | `pad.mod_beats` | 0.5 .. 64 beats | 16 |
| Mod Feedback | `pad.mod_feedback` | -0.9 .. 0.9 | 0.5 |

## TRANCE GATE

| Control | Key | Range | Default |
|---|---|---|---|
| Trance Gate | `pad.gate` | off / on | off |
| Gate Pattern | `pad.gate_pattern` | Sixteenths \| Eighths \| Rolling \| Gallop \| 3-3-2 \| Triplets | Sixteenths |
| Gate Depth | `pad.gate_depth` | 0 .. 1 | 0.85 |
| Gate Duty | `pad.gate_duty` | 0.05 .. 1 | 0.5 |
| Gate Attack | `pad.gate_attack` | 0.5 .. 60 ms | 3 |
| Gate Release | `pad.gate_release` | 0.5 .. 120 ms | 12 |
| Gate Tone | `pad.gate_tone` | 0 .. 1 | 0.4 |

## DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | `pad.delay_send` | 0 .. 1 | 0 |
| Delay Feedback | `pad.delay_feedback` | 0 .. 0.9 | 0.4 |
| Delay Left | `pad.delay_left` | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/16 |
| Delay Right | `pad.delay_right` | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/4 |
| Delay High Pass | `pad.delay_high_pass` | 150 .. 2000 Hz | 350 |
| Delay Low Pass | `pad.delay_low_pass` | 800 .. 16000 Hz | 6000 |

## SPACE & IMAGE

| Control | Key | Range | Default |
|---|---|---|---|
| Distance | `pad.distance` | 0 .. 1 | 1 |
| Width | `pad.width` | 0 .. 1 | 1 |
| Room Send | `pad.room_send` | 0 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Plate Send | `pad.plate_send` | 0 .. 1 | 0.35 |
| Hall Send | `pad.hall_send` | 0 .. 1 | 0 |
| Hall Gate | `pad.hall_gate` | off / on | off |
| Pan | `pad.pan` | -1 .. 1 | 0 |
| Slow Movement | `pad.slow_mod` | 0 .. 1 | 0.5 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `pad.level` | -36 .. 6 dB | -16 |
| Duck | `pad.duck` | 0 .. 1 | 0.5 |

## 14. Drone

A long, slowly moving tone under the breakdowns and the intros: the sound that holds a track together when the kick is gone. It is high-passed above the bass so it never takes the floor. It holds the tonic, its fifth and its octave, and like the pad never a minor ninth over the bass -- in a DJ blend a note of the incoming track that would stand a semitone over the outgoing tonic waits for the hand-over. With Bass Follows Chords on it follows the bass to the pad's root.




The Drone tab.

## SOUND

| Control | Key | Range | Default |
|---|---|---|---|
| Drone Own Sound | `mix.drone_own` | off / on | off |

## OSCILLATOR

| Control | Key | Range |
|---|---|---|
| Oscillator | `drone.osc` | Supersaw \| VA \| FM \| Wavetable |
| Table | `drone.table` | Classic \| Vocal \| Glass \| PWM \| Sync \| Formant Saw \| WaveEdit Hienharm \| WaveEdit Junox_ho \| WaveEdit Euclidea \| WaveEdit Sohler49 \| WaveEdit Vocal_fo \| WaveE |
| Position | `drone.position` | 0 .. 1 |
| Detune | `drone.detune` | 0 .. 1 |
| Mix | `drone.mix` | 0 .. 1 |
| Wave | `drone.wave` | 0 .. 1 |
| Dynamic Detune | `drone.dynamic_detune` | 0 .. 1 |
| Pulse Width | `drone.pulse_width` | 0.05 .. 0.95 |
| Drift | `drone.drift` | 0 .. 8 ct |

## WAVETABLE MOTION

| Control | Key | Range | Default |
|---|---|---|---|
| Position Env | `drone.pos_env` | -1 .. 1 | 0 |
| Position LFO | `drone.pos_lfo_depth` | 0 .. 0.5 | 0.1 |
| Position Decay | `drone.pos_decay` | 10 .. 8000 ms | 1500 |
| LFO Period | `drone.pos_lfo_beats` | 0.25 .. 64 beats | 32 |

## FM

| Control | Key | Range | Default |
|---|---|---|---|
| FM Ratio | `drone.fm_ratio` | 0.5 .. 8 | 2 |
| FM Index | `drone.fm_index` | 0 .. 10 | 2.5 |
| FM Decay | `drone.fm_decay` | 5 .. 2000 ms | 250 |

## SECOND OSCILLATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Oscillator 2 | `drone.osc2` | Off \| Supersaw \| VA \| FM \| Wavetable | Off |
| Osc 2 Mix | `drone.osc2_mix` | 0 .. 1 | 0.36 |
| Osc 2 Interval | `drone.osc2_interval` | -2 Oct \| -1 Oct \| -5th \| Unison \| +5th \| +1 Oct | -1 Oct |
| Osc 2 Detune | `drone.osc2_detune` | -50 .. 50 ct | 5 |

## FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `drone.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `drone.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `drone.cutoff` | 200 .. 18000 Hz | 900 |
| Resonance | `drone.resonance` | 0 .. 1 | 0.1 |
| Env Amount | `drone.env_amount` | 0 .. 6 oct | 0 |
| Filter Mode | `drone.filter_mode` | 0 .. 1 | 0 |
| Key Track | `drone.key_track` | 0 .. 1 | 0.5 |
| HP Floor | `drone.hp_floor` | 40 .. 400 Hz | 140 |
| HP Track | `drone.hp_track` | 0 .. 1 x f0 | 0.5 |
| Disperse | `drone.disperse` | 0 .. 8 x | 0 |
| Disperse Freq | `drone.disperse_freq` | 200 .. 8000 Hz | 1250 |

## FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `drone.filt_attack` | 0.1 .. 4000 ms | 0.1 |

| | | | |
|---|---|---|---|
| Filter Decay | drone.filter_decay | 5 .. 3000 ms | 400 |
| Filter Sustain | drone.filt_sustain | 0 .. 1 | 0 |
| Filter Release | drone.filt_release | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | drone.amp_attack | 0.3 .. 2000 ms | 1500 |
| Decay | drone.amp_decay | 5 .. 4000 ms | 2000 |
| Sustain | drone.amp_sustain | 0 .. 1 | 1 |
| Release | drone.amp_release | 5 .. 4000 ms | 2500 |
| Velocity | drone.vel_sens | 0 .. 1 | 0.4 |
| Glide | drone.glide | 0 .. 400 ms | 0 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | drone.menv_attack | 0.1 .. 8000 ms | 10 |
| Mod Decay | drone.menv_decay | 5 .. 12000 ms | 800 |
| Mod Sustain | drone.menv_sustain | 0 .. 1 | 0 |
| Mod Release | drone.menv_release | 5 .. 12000 ms | 400 |

### LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | drone.lfo1_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | drone.lfo1_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | drone.lfo1_rate | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | drone.lfo1_retrig | off / on | off |
| LFO 1 Fade | drone.lfo1_fade | 0 .. 8 s | 0 |

### LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | drone.lfo2_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | drone.lfo2_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | drone.lfo2_rate | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | drone.lfo2_retrig | off / on | off |
| LFO 2 Fade | drone.lfo2_fade | 0 .. 8 s | 0 |

### LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | drone.lfo3_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | drone.lfo3_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | drone.lfo3_rate | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | drone.lfo3_retrig | off / on | off |
| LFO 3 Fade | drone.lfo3_fade | 0 .. 8 s | 0 |

### LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | drone.lfo4_shape | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | drone.lfo4_sync | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | drone.lfo4_rate | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | drone.lfo4_retrig | off / on | off |
| LFO 4 Fade | drone.lfo4_fade | 0 .. 8 s | 0 |

### MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | drone.mx1_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | drone.mx1_dst | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 1 Amount | drone.mx1_amount | -1 .. 1 | 0 |
| Mod 2 Source | drone.mx2_src | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |

| | | | |
|---|---|---|---|
| Mod 2 Target | `drone.mx2_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 2 Amount | `drone.mx2_amount` | -1 .. 1 | 0 |
| Mod 3 Source | `drone.mx3_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | `drone.mx3_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 3 Amount | `drone.mx3_amount` | -1 .. 1 | 0 |
| Mod 4 Source | `drone.mx4_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | `drone.mx4_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 4 Amount | `drone.mx4_amount` | -1 .. 1 | 0 |
| Mod 5 Source | `drone.mx5_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | `drone.mx5_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 5 Amount | `drone.mx5_amount` | -1 .. 1 | 0 |
| Mod 6 Source | `drone.mx6_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | `drone.mx6_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 6 Amount | `drone.mx6_amount` | -1 .. 1 | 0 |
| Mod 7 Source | `drone.mx7_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | `drone.mx7_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 7 Amount | `drone.mx7_amount` | -1 .. 1 | 0 |
| Mod 8 Source | `drone.mx8_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | `drone.mx8_dst` | Off \| Pitch \| Osc 2 Pitch \| Pulse Width \| Table Position \| FM Index \| Cutoff \| Resonance \| Filter Mode \| Level \| Pan | Off |
| Mod 8 Amount | `drone.mx8_amount` | -1 .. 1 | 0 |

### VIBRATO & TREMOLO

| Control | Key | Range | Default |
|---|---|---|---|
| Voice LFO | `drone.lfo_beats` | 0.25 .. 64 beats | 32 |
| LFO to Cutoff | `drone.lfo_cutoff` | 0 .. 4 oct | 0.32 |
| LFO to Pitch | `drone.lfo_pitch` | 0 .. 50 ct | 0 |
| LFO to Level | `drone.lfo_amp` | 0 .. 1 | 0.08 |

### VOICE FX

| Control | Key | Range | Default |
|---|---|---|---|
| Modulation | `drone.mod` | Off \| Flanger \| Phaser \| Comb | Off |
| Mod Mix | `drone.mod_mix` | 0 .. 1 | 0 |
| Mod Depth | `drone.mod_depth` | 0 .. 1 | 0.7 |
| Mod Period | `drone.mod_beats` | 0.5 .. 64 beats | 16 |
| Mod Feedback | `drone.mod_feedback` | -0.9 .. 0.9 | 0.5 |

### TRANCE GATE

| Control | Key | Range | Default |
|---|---|---|---|
| Trance Gate | `drone.gate` | off / on | off |
| Gate Pattern | `drone.gate_pattern` | Sixteenths \| Eighths \| Rolling \| Gallop \| 3-3-2 \| Triplets | Sixteenths |
| Gate Depth | `drone.gate_depth` | 0 .. 1 | 0.85 |
| Gate Duty | `drone.gate_duty` | 0.05 .. 1 | 0.5 |
| Gate Attack | `drone.gate_attack` | 0.5 .. 60 ms | 3 |
| Gate Release | `drone.gate_release` | 0.5 .. 120 ms | 12 |
| Gate Tone | `drone.gate_tone` | 0 .. 1 | 0.4 |

### DELAY

| Control | Key | Range | Default |
|---|---|---|---|
| Delay Send | `drone.delay_send` | 0 .. 1 | 0 |
| Delay Feedback | `drone.delay_feedback` | 0 .. 0.9 | 0.4 |
| Delay Left | `drone.delay_left` | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/16 |
| Delay Right | `drone.delay_right` | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 1/4 |
| Delay High Pass | `drone.delay_high_pass` | 150 .. 2000 Hz | 350 |
| Delay Low Pass | `drone.delay_low_pass` | 800 .. 16000 Hz | 6000 |

### SPACE & IMAGE

| Control | Key | Range | Default |
|---|---|---|---|
| Distance | `drone.distance` | 0 .. 1 | 1 |

| | | | |
|---|---|---|---|
| Width | `drone.width` | 0 .. 1 | 1 |
| Room Send | `drone.room_send` | 0 .. 1 | 0 |
| Plate Send | `drone.plate_send` | 0 .. 1 | 0.5 |
| Hall Send | `drone.hall_send` | 0 .. 1 | 0 |
| Hall Gate | `drone.hall_gate` | off / on | off |
| Pan | `drone.pan` | -1 .. 1 | 0 |
| Slow Movement | `drone.slow_mod` | 0 .. 1 | 0.25 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `drone.level` | -36 .. 6 dB | -17 |
| Duck | `drone.duck` | 0 .. 1 | 0.45 |

## 15. SFX / FX

The effect generator makes risers, downlifters, impacts, sweeps, formant shots, reverse swells and zaps; the composer places them at the form's boundaries -- a riser over the last eight bars of a buildup, ending exactly on the drop, the formant shot on the last beat of the pre-drop break, a descending sweep into the drop.

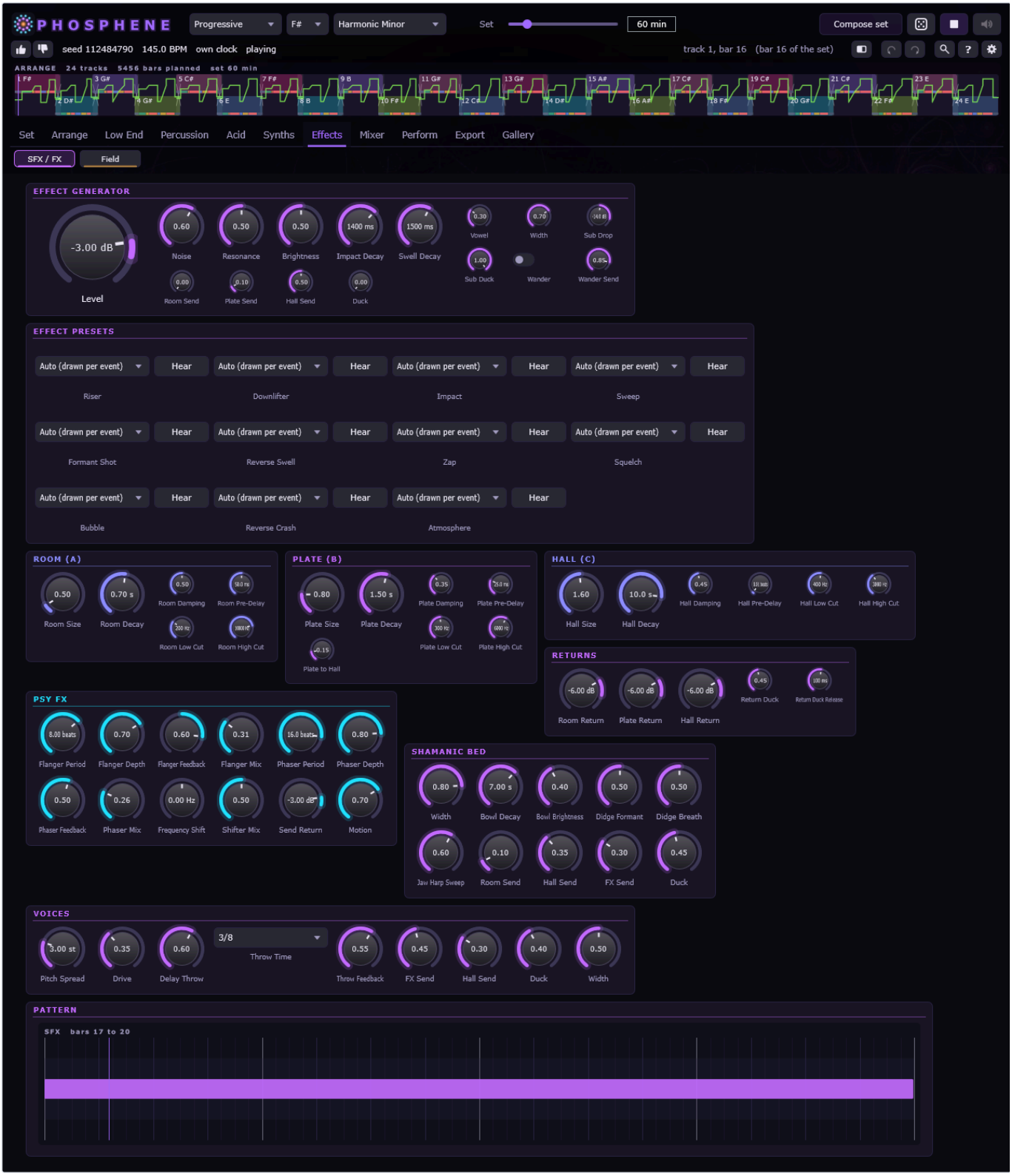
Short one-shots -- zaps, squelches, bubbles, alien chatter -- fall at the ends of phrases, about one in eight bars of a groove and one in four of the climax; a groove is never left eight bars without an effect, and what fills such a gap is a sweep or a swell, never a zap.

Effect Presets: every family of the effect bank (2048 presets in eleven families: riser, downlifter, impact, sweep, formant shot, reverse swell, zap, squelch, bubble, reverse crash, atmosphere) has a chooser. Auto lets the generator draw a preset of the family for every event, never the same twice in a track; a number makes every event of that family play that preset. "Hear" plays the chosen preset in the running set (with Auto, one of the family at random); while the transport stands nothing sounds.

Space Shot: a recording of NASA's, up to 25 seconds -- a rocket launch, Sputnik's beep, Saturn's radio emissions, a quake on Mars -- played whole, once, on this strip: after the cut of a breakdown, now and then in a long intro, most in Full-On and Hi-Tech, and never within two bars of a spoken phrase. It comes from the field recordings (Field tab); without them it stays silent.

The tonal layer of an effect -- an atmosphere's held tone above all -- is mapped into the mode, and never onto the b2 or the leading tone: a minor seventh the mode lacks becomes the fifth, not the major seventh.

The two send effects are a short room and a long hall, both feedback delay networks. Their returns are ducked with the kick like everything else, and high-passed: a reverb tail in the bass register is mud.




The SFX / FX tab.

## EFFECT GENERATOR

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `sfx.level` | -36 .. 6 dB | -3 |
| Noise | `sfx.noise` | 0 .. 1 | 0.6 |
| Resonance | `sfx.resonance` | 0 .. 1 | 0.5 |
| Brightness | `sfx.brightness` | 0 .. 1 | 0.5 |
| Impact Decay | `sfx.impact_decay` | 200 .. 4000 ms | 1400 |
| Swell Decay | `sfx.swell_decay` | 200 .. 6000 ms | 1500 |
| Vowel | `sfx.vowel` | 0 .. 1 | 0.3 |
| Width | `sfx.width` | 0 .. 1 | 0.7 |

| | | | |
|---|---|---|---|
| Sub Drop | `sfx.sub_level` | -36 .. 6 dB | -14 |
| Sub Duck | `sfx.sub_duck` | 0 .. 1 | 1 |
| Wander | `sfx.wander` | off / on | off |
| Wander Send | `sfx.wander_send` | 0 .. 1 | 0.85 |
| Room Send | `sfx.room_send` | 0 .. 1 | 0 |
| Plate Send | `sfx.plate_send` | 0 .. 1 | 0.1 |
| Hall Send | `sfx.hall_send` | 0 .. 1 | 0.5 |
| Duck | `sfx.duck` | 0 .. 1 | 0 |

### EFFECT PRESETS

| Control | Key | Range | Default |
|---|---|---|---|
| Riser | `sfx.preset_riser` | 0 .. 256 | 0 |
| Downlifter | `sfx.preset_downlifter` | 0 .. 128 | 0 |
| Impact | `sfx.preset_impact` | 0 .. 128 | 0 |
| Sweep | `sfx.preset_sweep` | 0 .. 256 | 0 |
| Formant Shot | `sfx.preset_formant_shot` | 0 .. 96 | 0 |
| Reverse Swell | `sfx.preset_reverse_swell` | 0 .. 256 | 0 |
| Zap | `sfx.preset_zap` | 0 .. 96 | 0 |
| Squelch | `sfx.preset_squelch` | 0 .. 128 | 0 |
| Bubble | `sfx.preset_bubble` | 0 .. 64 | 0 |
| Reverse Crash | `sfx.preset_reverse_crash` | 0 .. 128 | 0 |
| Atmosphere | `sfx.preset_atmosphere` | 0 .. 512 | 0 |

### ROOM (A)

| Control | Key | Range | Default |
|---|---|---|---|
| Room Size | `fx.room_size` | 0.3 .. 3 | 0.5 |
| Room Decay | `fx.room_decay` | 0.1 .. 4 s | 0.7 |
| Room Damping | `fx.room_damping` | 0 .. 1 | 0.5 |
| Room Pre-Delay | `fx.room_pre_delay` | 0 .. 100 ms | 50 |
| Room Low Cut | `fx.room_low_cut` | 100 .. 1000 Hz | 200 |
| Room High Cut | `fx.room_high_cut` | 1000 .. 20000 Hz | 10000 |

### PLATE (B)

| Control | Key | Range | Default |
|---|---|---|---|
| Plate Size | `fx.plate_size` | 0.3 .. 3 | 0.8 |
| Plate Decay | `fx.plate_decay` | 0.2 .. 8 s | 1.5 |
| Plate Damping | `fx.plate_damping` | 0 .. 1 | 0.35 |
| Plate Pre-Delay | `fx.plate_pre_delay` | 0 .. 100 ms | 25 |
| Plate Low Cut | `fx.plate_low_cut` | 100 .. 1000 Hz | 300 |
| Plate High Cut | `fx.plate_high_cut` | 1000 .. 20000 Hz | 6000 |
| Plate to Hall | `fx.plate_to_hall` | 0 .. 1 | 0.15 |

### HALL (C)

| Control | Key | Range | Default |
|---|---|---|---|
| Hall Size | `fx.hall_size` | 0.3 .. 3 | 1.6 |
| Hall Decay | `fx.hall_decay` | 0.3 .. 20 s | 10 |
| Hall Damping | `fx.hall_damping` | 0 .. 1 | 0.45 |
| Hall Pre-Delay | `fx.hall_pre_delay` | 0 .. 0.5 beats | 0.0125 |
| Hall Low Cut | `fx.low_cut` | 150 .. 1000 Hz | 400 |
| Hall High Cut | `fx.high_cut` | 1000 .. 20000 Hz | 3000 |

### RETURNS

| Control | Key | Range | Default |
|---|---|---|---|
| Room Return | `fx.room_return` | -36 .. 6 dB | -6 |
| Plate Return | `fx.plate_return` | -36 .. 6 dB | -6 |
| Hall Return | `fx.hall_return` | -36 .. 6 dB | -6 |
| Return Duck | `fx.return_duck` | 0 .. 1 | 0.45 |
| Return Duck Release | `fx.return_duck_release` | 10 .. 800 ms | 100 |

## PSY FX

| Control | Key | Range | Default |
|---|---|---|---|
| Flanger Period | `psyfx.flanger_beats` | 0.5 .. 32 beats | 8 |
| Flanger Depth | `psyfx.flanger_depth` | 0 .. 1 | 0.7 |
| Flanger Feedback | `psyfx.flanger_feedback` | -0.9 .. 0.9 | 0.6 |
| Flanger Mix | `psyfx.flanger_mix` | 0 .. 1 | 0.35 |
| Phaser Period | `psyfx.phaser_beats` | 0.5 .. 32 beats | 16 |
| Phaser Depth | `psyfx.phaser_depth` | 0 .. 1 | 0.8 |
| Phaser Feedback | `psyfx.phaser_feedback` | 0 .. 0.9 | 0.5 |
| Phaser Mix | `psyfx.phaser_mix` | 0 .. 1 | 0.3 |
| Frequency Shift | `psyfx.shift_hz` | -500 .. 500 Hz | 0 |
| Shifter Mix | `psyfx.shift_mix` | 0 .. 1 | 0.5 |
| Send Return | `psyfx.return` | -36 .. 6 dB | -3 |
| Motion | `psyfx.motion` | 0 .. 1 | 0.7 |

## SHAMANIC BED

| Control | Key | Range | Default |
|---|---|---|---|
| Width | `texture.width` | 0 .. 1 | 0.8 |
| Bowl Decay | `texture.bowl_decay` | 1 .. 20 s | 7 |
| Bowl Brightness | `texture.bowl_bright` | 0 .. 1 | 0.4 |
| Didge Formant | `texture.didge_formant` | 0 .. 1 | 0.5 |
| Didge Breath | `texture.didge_breath` | 0 .. 1 | 0.5 |
| Jaw Harp Sweep | `texture.jaw_sweep` | 0 .. 1 | 0.6 |
| Room Send | `texture.room_send` | 0 .. 1 | 0.1 |
| Hall Send | `texture.hall_send` | 0 .. 1 | 0.35 |
| FX Send | `texture.fx_send` | 0 .. 1 | 0.3 |
| Duck | `texture.duck` | 0 .. 1 | 0.45 |

## VOICES

| Control | Key | Range | Default |
|---|---|---|---|
| Pitch Spread | `vocal.pitch` | 0 .. 12 st | 3 |
| Drive | `vocal.drive` | 0 .. 1 | 0.35 |
| Delay Throw | `vocal.throw_send` | 0 .. 1 | 0.6 |
| Throw Time | `vocal.throw_beats` | 1/16 \| 1/8 \| 3/16 \| 1/4 \| 3/8 \| 1/2 | 3/8 |
| Throw Feedback | `vocal.throw_feedback` | 0 .. 0.9 | 0.55 |
| FX Send | `vocal.fx_send` | 0 .. 1 | 0.45 |
| Hall Send | `vocal.hall_send` | 0 .. 1 | 0.3 |
| Duck | `vocal.duck` | 0 .. 1 | 0.4 |
| Width | `vocal.width` | 0 .. 1 | 0.5 |

## 16. Field

The Field track plays field recordings: a place under the music -- a rainforest, a night full of insects, rain on a roof, a river, a cave, a machine hall, the wind on Mars. It is a sampler, and its recordings are the original files, not converted: 120 recordings in 30 categories (four each, the steadiest and most different of their category; generated with Stable Audio 3 for Noctuary) and NASA's sounds from space. They are an optional part of the installation (2 GB). Without them everything plays; only this track and the NASA shots are silent, and the Library line says so.

Where it plays: under the intro from its first bar (so the place changes with the track, over the DJ's blend), over a breakdown, in the outro up to the blend, and -- in Dark Forest, rarely in Progressive -- sixteen quiet bars under a long groove. Never in a build-up or a drop. How often follows the style: Dark Forest always has its place, Hi-Tech least. compose.field_density scales it.

What plays: with A Category on Auto the composer draws a preset for every track by its style -- forest, night, insects and swamps for Dark Forest, sea, fire and ritual for Goa, wind, rain and water for Progressive, machines and radio for Hi-Tech, space for Full-On -- and sends it as it does the synths' presets: the knobs stay as they are. Choose a category and the knobs play, in every track.

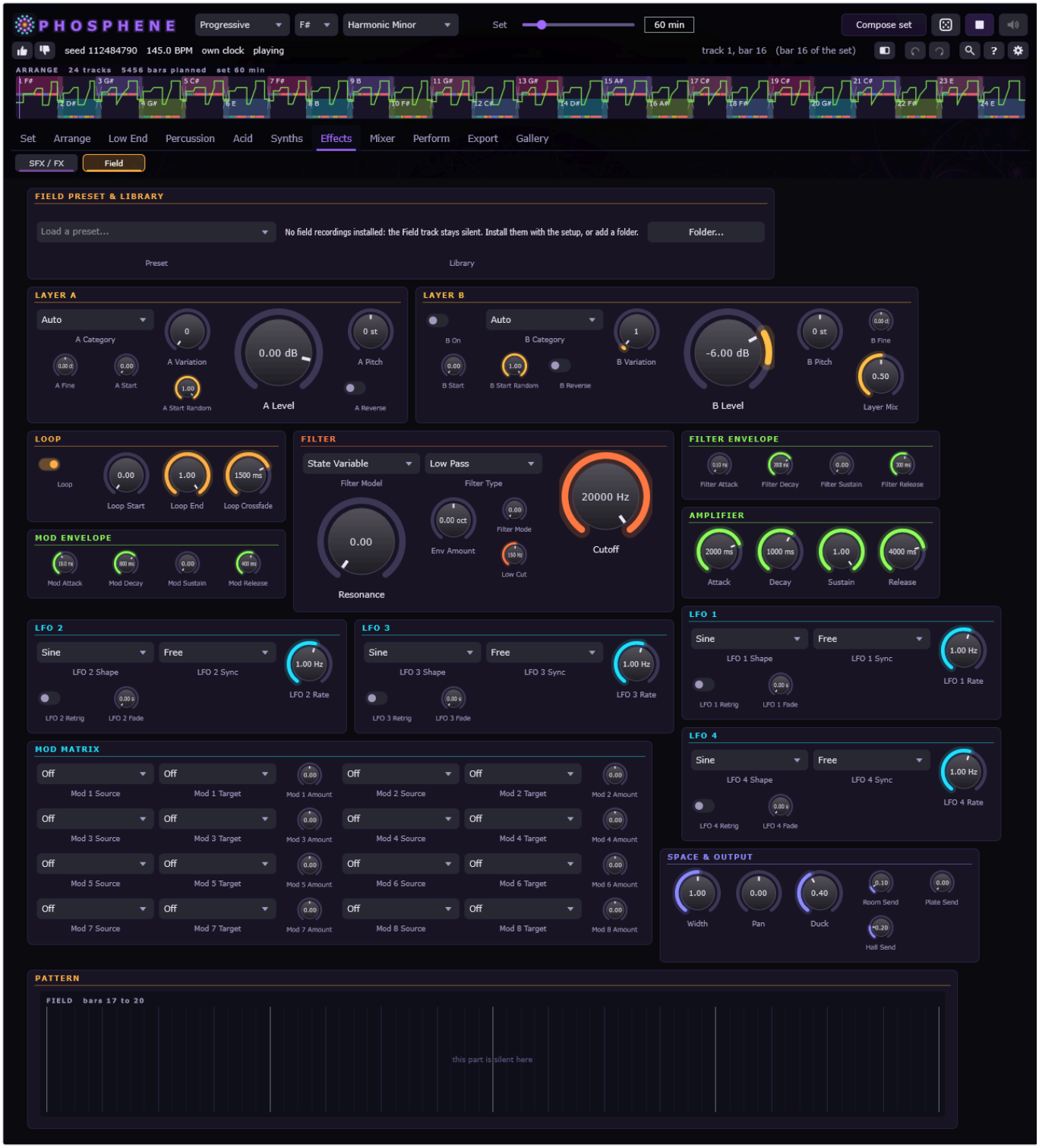
Preset: loads one of the factory presets into the knobs -- one per recording (by category) and the scenes, two recordings as one place (Jungle Night: a night and its insects; Swamp: the wetland and its mud). The category is then no longer Auto; set it back to hand the choice to the composer again. Library: how many recordings were found, and where. Folder...: a folder of your own recordings beside the installed ones -- FLAC files named fr-<category>-<anything>.flac (Noctuary's whole library of 2938 fits as it is); cancel the dialog to remove it.

Layer A and B: each a recording, chosen by category and variation (the variations of a category are its files in name order), with its level, its pitch (semitones and cents, by resampling), where it starts and how far that start may wander per event, and its direction. B is off by default; Layer Mix weighs the two (the middle plays both whole).

Loop: over any stretch of the recording (Loop Start to Loop End), joined by a crossfade of its own length -- over the last stretch before the loop's end a second read head runs from the loop's start, and the two are mixed with equal power, because two stretches of a field recording are uncorrelated and keep their loudness only so. What is heard at the seam is always a blend, never a cut, so a recording that was not made as a loop plays as one. Without the loop a layer ends with its recording.

Filter, Filter Envelope, Amplifier and the modulation block are the synths' (Filters and modulation); the matrix reaches pitch, cutoff, resonance, the filter's mode, level and pan. The Low Cut (24 dB per octave) keeps the recordings out of the range that belongs to kick and bass -- wind and water have most of their power there. The long amp envelope (2 s in, 4 s out) is how a place enters and leaves.

The recordings are loaded when a track is planned, a track ahead of their first note, and kept in memory; an offline render waits for them, so it is the same every time.




The Field tab.

## LAYER A

| Control | Key | Range |
|---|---|---|
| A Category | `field.a_category` | Auto \| Rainforest \| Night Country \| Insects \| Foliage \| Wetland \| Mud Bubbles \| Cave \| River \| Sea \| Rain on Land \| Rain on Roof \| Wind \| Underwater \| Ice and Snow \| |
| A Variation | `field.a_variation` | 0 .. 63 |
| A Level | `field.a_level` | -36 .. 6 dB |
| A Pitch | `field.a_pitch` | -24 .. 24 st |
| A Fine | `field.a_fine` | -100 .. 100 ct |
| A Start | `field.a_start` | 0 .. 1 |
| A Start Random | `field.a_start_random` | 0 .. 1 |

| | | |
|---|---|---|
| A Reverse | `field.a_reverse` | off / on |

### LAYER B

| Control | Key | Range |
|---|---|---|
| B On | `field.b_on` | off / on |
| B Category | `field.b_category` | Auto \| Rainforest \| Night Country \| Insects \| Foliage \| Wetland \| Mud Bubbles \| Cave \| River \| Sea \| Rain on Land \| Rain on Roof \| Wind \| Underwater \| Ice and Snow \| |
| B Variation | `field.b_variation` | 0 .. 63 |
| B Level | `field.b_level` | -36 .. 6 dB |
| B Pitch | `field.b_pitch` | -24 .. 24 st |
| B Fine | `field.b_fine` | -100 .. 100 ct |
| B Start | `field.b_start` | 0 .. 1 |
| B Start Random | `field.b_start_random` | 0 .. 1 |
| B Reverse | `field.b_reverse` | off / on |
| Layer Mix | `field.layer_mix` | 0 .. 1 |

### LOOP

| Control | Key | Range | Default |
|---|---|---|---|
| Loop | `field.loop` | off / on | on |
| Loop Start | `field.loop_start` | 0 .. 1 | 0 |
| Loop End | `field.loop_end` | 0 .. 1 | 1 |
| Loop Crossfade | `field.loop_xfade` | 10 .. 10000 ms | 1500 |

### FILTER

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Model | `field.filter_model` | State Variable \| Moog Ladder \| Prophet (SSM2040) \| Juno (IR3109) \| Oberheim SEM \| Xpander \| Diode Ladder \| Korg35 (MS-20) \| Polivoks \| Wasp | State Variable |
| Filter Type | `field.filter_type` | Low Pass \| Band Pass \| High Pass \| Notch | Low Pass |
| Cutoff | `field.cutoff` | 40 .. 20000 Hz | 20000 |
| Resonance | `field.resonance` | 0 .. 1 | 0 |
| Env Amount | `field.env_amount` | -8 .. 8 oct | 0 |
| Filter Mode | `field.filter_mode` | 0 .. 1 | 0 |
| Low Cut | `field.low_cut` | 20 .. 1000 Hz | 150 |

### FILTER ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Filter Attack | `field.filt_attack` | 0.1 .. 4000 ms | 0.1 |
| Filter Decay | `field.filter_decay` | 5 .. 30000 ms | 2000 |
| Filter Sustain | `field.filt_sustain` | 0 .. 1 | 0 |
| Filter Release | `field.filt_release` | 5 .. 8000 ms | 300 |

### AMPLIFIER

| Control | Key | Range | Default |
|---|---|---|---|
| Attack | `field.amp_attack` | 1 .. 30000 ms | 2000 |
| Decay | `field.amp_decay` | 5 .. 30000 ms | 1000 |
| Sustain | `field.amp_sustain` | 0 .. 1 | 1 |
| Release | `field.amp_release` | 5 .. 30000 ms | 4000 |

### MOD ENVELOPE

| Control | Key | Range | Default |
|---|---|---|---|
| Mod Attack | `field.menv_attack` | 0.1 .. 8000 ms | 10 |
| Mod Decay | `field.menv_decay` | 5 .. 12000 ms | 800 |
| Mod Sustain | `field.menv_sustain` | 0 .. 1 | 0 |
| Mod Release | `field.menv_release` | 5 .. 12000 ms | 400 |

## LFO 1

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 1 Shape | `field.lfo1_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 1 Sync | `field.lfo1_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 1 Rate | `field.lfo1_rate` | 0.01 .. 40 Hz | 1 |
| LFO 1 Retrig | `field.lfo1_retrig` | off / on | off |
| LFO 1 Fade | `field.lfo1_fade` | 0 .. 8 s | 0 |

## LFO 2

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 2 Shape | `field.lfo2_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 2 Sync | `field.lfo2_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 2 Rate | `field.lfo2_rate` | 0.01 .. 40 Hz | 1 |
| LFO 2 Retrig | `field.lfo2_retrig` | off / on | off |
| LFO 2 Fade | `field.lfo2_fade` | 0 .. 8 s | 0 |

## LFO 3

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 3 Shape | `field.lfo3_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 3 Sync | `field.lfo3_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 3 Rate | `field.lfo3_rate` | 0.01 .. 40 Hz | 1 |
| LFO 3 Retrig | `field.lfo3_retrig` | off / on | off |
| LFO 3 Fade | `field.lfo3_fade` | 0 .. 8 s | 0 |

## LFO 4

| Control | Key | Range | Default |
|---|---|---|---|
| LFO 4 Shape | `field.lfo4_shape` | Sine \| Triangle \| Saw Up \| Saw Down \| Square \| Sample & Hold \| Smooth Random | Sine |
| LFO 4 Sync | `field.lfo4_sync` | Free \| 4 Bars \| 2 Bars \| 1 Bar \| 1/2 \| 1/4 \| 1/8 \| 1/16 \| 1/4 T \| 1/8 T | Free |
| LFO 4 Rate | `field.lfo4_rate` | 0.01 .. 40 Hz | 1 |
| LFO 4 Retrig | `field.lfo4_retrig` | off / on | off |
| LFO 4 Fade | `field.lfo4_fade` | 0 .. 8 s | 0 |

## MOD MATRIX

| Control | Key | Range | Default |
|---|---|---|---|
| Mod 1 Source | `field.mx1_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 1 Target | `field.mx1_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 1 Amount | `field.mx1_amount` | -1 .. 1 | 0 |
| Mod 2 Source | `field.mx2_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 2 Target | `field.mx2_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 2 Amount | `field.mx2_amount` | -1 .. 1 | 0 |
| Mod 3 Source | `field.mx3_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 3 Target | `field.mx3_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 3 Amount | `field.mx3_amount` | -1 .. 1 | 0 |
| Mod 4 Source | `field.mx4_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 4 Target | `field.mx4_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 4 Amount | `field.mx4_amount` | -1 .. 1 | 0 |
| Mod 5 Source | `field.mx5_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 5 Target | `field.mx5_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 5 Amount | `field.mx5_amount` | -1 .. 1 | 0 |
| Mod 6 Source | `field.mx6_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 6 Target | `field.mx6_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 6 Amount | `field.mx6_amount` | -1 .. 1 | 0 |
| Mod 7 Source | `field.mx7_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 7 Target | `field.mx7_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |
| Mod 7 Amount | `field.mx7_amount` | -1 .. 1 | 0 |
| Mod 8 Source | `field.mx8_src` | Off \| LFO 1 \| LFO 2 \| LFO 3 \| LFO 4 \| Mod Env \| Filter Env \| Velocity \| Key \| Random | Off |
| Mod 8 Target | `field.mx8_dst` | Off \| Pulse Width \| Cutoff \| Resonance \| Filter Mode \| Level | Off |

| | | | |
|---|---|---|---|
| Mod 8 Amount | `field.mx8_amount` | -1 .. 1 | 0 |

### SPACE & OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Width | `field.width` | 0 .. 2 | 1 |
| Pan | `field.pan` | -1 .. 1 | 0 |
| Duck | `field.duck` | 0 .. 1 | 0.4 |
| Room Send | `field.room_send` | 0 .. 1 | 0.1 |
| Plate Send | `field.plate_send` | 0 .. 1 | 0 |
| Hall Send | `field.hall_send` | 0 .. 1 | 0.2 |

## 17. Mixer / Master

A channel strip per part: its pan (the six voices), its sends to the room, the hall and its delay or effects bus, its duck under the kick, the mute ("M") and the fader, beside a meter of what the part puts into the mix -- after fader, trance gate and duck, before the master. The green-to-red bar is the RMS level, the line above it the peak, held for a second and a half; the figure under the meter is that peak in dBFS. The knobs of a strip are the same parameters its synth page shows. Kick and bass have no strip level of their own: their fader is the synth's level.

The fader stays where you put it. Where the composer plays a part at another level -- the level match that holds every track's parts to one loudness, a preset's level correction -- a bright tick beside the fader shows the level that plays, joined to the fader's cap by a line.

Beside the strips the track gain and the sidechain's times, and the master chain: bus compressor, mono bass, a two-times oversampled knee clipper, a lookahead true-peak limiter and a safety clip, metered to ITU-R BS.1770.

Auto Gain renders four four-bar windows of the track through the whole master chain, measures them and offsets the master gain so that the output meets the loudness target. It runs when a track is planned, so it uses the knobs as they stand at that moment.

Depth comes in three planes, and three rooms. The front -- kick, bass and the lead -- is dry, the lead with a little of the near room (A: short, early reflections, a 50 ms pre-delay). The middle -- acid, counter, arp and stab -- sends to the plate (B, the main room: 1.5 s in a drop, 4 s in a breakdown, 25 ms pre-delay). The back -- pads and drone -- sends to the plate as well, more of it, and the plate in turn feeds the hall (C, the far room: ten seconds, 400 Hz .. 3 kHz), which the bed and the voices reach directly. The far room is heard only in intros and breakdowns: in a drop there are only A and B. Pre-delays are distance cues, not tempo values: near sources get the long one (their direct sound arrives well before the first reflection), far ones almost none. Each room has its own send filter (the Rooms group on the FX page).

Every voice carries a Distance (0 near .. 1 far, its own plane by default: the lead 0, the middle 0.5, pad and drone 1). Moving it moves all the cues of distance at once -- level, a low pass, the plate send and the width -- so a voice is never near by one cue and far by another. The composer uses it: the lead steps back in a breakdown and comes forward in the drop, and in the main breakdown the counter-lead arrives from far away over sixteen bars. Pads and drone also move slowly on their own (Slow Movement: cutoff, colour and width on periods of 13, 21 and 34 seconds, in golden-mean ratios, so no pattern is heard to come round).

Every strip gives way to the kick by what its plane needs: the bass 4 dB, the lines about 2 dB and back within 65 ms (Duck Release (Lines)), the back plane 5 to 6 dB within 125 ms (Duck Release), the returns 5 dB within 100 ms (Return Duck Release). The effects do not duck: an impact lands on the drop's kick. The lead drives two ducks of its own: the counter steps back while the lead plays (Counter under Lead), and so does the pad's presence band between 500 Hz and 3 kHz, for as long as a lead note sounds (Pad Band under Lead).

The composer mixes each phase differently. A breakdown stands 7 dB under the drop before it (after the master's compressor and limiter that is about 5 dB of loudness), with the pad at full width and a little louder, the lead further back and the far room open; a buildup narrows the pad, pulls the plate down to 1.2 seconds, fades the far room out and lifts the lines' high pass, and the drop opens it all again within a bar -- the lead at the front, the pad at its full width again, the pad and drone ducking deeper. In the intro and the breakdown, where pad and drone are often the only tonal voices, their unison stays within about six cents: held, a wider stack beats. While the bass plays, the pad leaves the third to the lead. On the last beat before every eight-bar change of a groove or a drop the effects are thrown into the plate for that one beat.

Monitor (Master page): Mono, Sub (a low pass at 80 Hz, where the kick-bass gap and the tuning are judged) or Side, after the meter -- a listening switch, it changes nothing the loudness readings see.




The Mixer / Master tab.

## CHANNELS

| Control | Key | Range | Default |
|---|---|---|---|
| Level | `kick.level` | -36 .. 6 dB | -6 |
| Kick Mute | `mix.kick_mute` | off / on | off |
| Level | `bass.level` | -36 .. 6 dB | -3 |
| Bass Mute | `mix.bass_mute` | off / on | off |
| Perc Level | `mix.perc_level` | -24 .. 12 dB | 1.2 |
| Perc Mute | `mix.perc_mute` | off / on | off |
| Perc Room | `mix.perc_room` | 0 .. 1 | 0.12 |
| Perc Hall | `mix.perc_hall` | 0 .. 1 | 0 |
| Acid Level | `mix.acid_level` | -24 .. 12 dB | 0 |
| Acid Mute | `mix.acid_mute` | off / on | off |
| Room Send | `acid.room_send` | 0 .. 1 | 0.1 |
| Hall Send | `acid.hall_send` | 0 .. 1 | 0 |
| Delay Send | `acid.delay_send` | 0 .. 1 | 0.25 |
| Duck | `acid.duck` | 0 .. 1 | 0.15 |
| Lead Level | `mix.lead_level` | -24 .. 12 dB | -6 |
| Lead Mute | `mix.lead_mute` | off / on | off |
| Pan | `lead.pan` | -1 .. 1 | -0.2 |
| Room Send | `lead.room_send` | 0 .. 1 | 0.1 |
| Hall Send | `lead.hall_send` | 0 .. 1 | 0 |
| Delay Send | `lead.delay_send` | 0 .. 1 | 0.3 |
| Duck | `lead.duck` | 0 .. 1 | 0.2 |
| Counter Level | `mix.counter_level` | -24 .. 12 dB | 5 |
| Counter Mute | `mix.counter_mute` | off / on | off |
| Pan | `counter.pan` | -1 .. 1 | 0.2 |
| Room Send | `counter.room_send` | 0 .. 1 | 0 |
| Hall Send | `counter.hall_send` | 0 .. 1 | 0 |

| | | | |
|---|---|---|---|
| Delay Send | `counter.delay_send` | 0 .. 1 | 0.4 |
| Duck | `counter.duck` | 0 .. 1 | 0.2 |
| Arp Level | `mix.arp_level` | -24 .. 12 dB | -5 |
| Arp Mute | `mix.arp_mute` | off / on | off |
| Pan | `arp.pan` | -1 .. 1 | 0 |
| Room Send | `arp.room_send` | 0 .. 1 | 0 |
| Hall Send | `arp.hall_send` | 0 .. 1 | 0 |
| Delay Send | `arp.delay_send` | 0 .. 1 | 0.35 |
| Duck | `arp.duck` | 0 .. 1 | 0.2 |
| Stab Level | `mix.stab_level` | -24 .. 12 dB | -3 |
| Stab Mute | `mix.stab_mute` | off / on | off |
| Pan | `stab.pan` | -1 .. 1 | 0 |
| Room Send | `stab.room_send` | 0 .. 1 | 0 |
| Hall Send | `stab.hall_send` | 0 .. 1 | 0 |
| Delay Send | `stab.delay_send` | 0 .. 1 | 0.45 |
| Duck | `stab.duck` | 0 .. 1 | 0.2 |
| Pad Level | `mix.pad_level` | -24 .. 12 dB | 0 |
| Pad Mute | `mix.pad_mute` | off / on | off |
| Pan | `pad.pan` | -1 .. 1 | 0 |
| Room Send | `pad.room_send` | 0 .. 1 | 0 |
| Hall Send | `pad.hall_send` | 0 .. 1 | 0 |
| Delay Send | `pad.delay_send` | 0 .. 1 | 0 |
| Duck | `pad.duck` | 0 .. 1 | 0.5 |
| Drone Level | `mix.drone_level` | -24 .. 12 dB | 0 |
| Drone Mute | `mix.drone_mute` | off / on | off |
| Pan | `drone.pan` | -1 .. 1 | 0 |
| Room Send | `drone.room_send` | 0 .. 1 | 0 |
| Hall Send | `drone.hall_send` | 0 .. 1 | 0 |
| Delay Send | `drone.delay_send` | 0 .. 1 | 0 |
| Duck | `drone.duck` | 0 .. 1 | 0.45 |
| SFX Level | `mix.sfx_level` | -24 .. 12 dB | 4.5 |
| SFX Mute | `mix.sfx_mute` | off / on | off |
| Room Send | `sfx.room_send` | 0 .. 1 | 0 |
| Hall Send | `sfx.hall_send` | 0 .. 1 | 0.5 |
| Duck | `sfx.duck` | 0 .. 1 | 0 |
| Texture Level | `mix.texture_level` | -24 .. 12 dB | 9 |
| Texture Mute | `mix.texture_mute` | off / on | off |
| Room Send | `texture.room_send` | 0 .. 1 | 0.1 |
| Hall Send | `texture.hall_send` | 0 .. 1 | 0.35 |
| FX Send | `texture.fx_send` | 0 .. 1 | 0.3 |
| Duck | `texture.duck` | 0 .. 1 | 0.45 |
| Vocal Level | `mix.vocal_level` | -24 .. 12 dB | 7 |
| Vocal Mute | `mix.vocal_mute` | off / on | off |
| Hall Send | `vocal.hall_send` | 0 .. 1 | 0.3 |
| FX Send | `vocal.fx_send` | 0 .. 1 | 0.45 |
| Delay Throw | `vocal.throw_send` | 0 .. 1 | 0.6 |
| Duck | `vocal.duck` | 0 .. 1 | 0.4 |
| Field Level | `mix.field_level` | -24 .. 12 dB | -14 |
| Field Mute | `mix.field_mute` | off / on | off |
| Pan | `field.pan` | -1 .. 1 | 0 |
| Room Send | `field.room_send` | 0 .. 1 | 0.1 |
| Hall Send | `field.hall_send` | 0 .. 1 | 0.2 |
| Duck | `field.duck` | 0 .. 1 | 0.4 |

### TRACK & SIDECHAIN

| Control | Key | Range | Default |
|---|---|---|---|
| Track Gain | `mix.track_gain` | -12 .. 12 dB | 0 |

| | | | |
|---|---|---|---|
| Duck Attack | `mix.duck_attack` | 0.5 .. 30 ms | 2 |
| Duck Hold | `mix.duck_hold` | 0 .. 200 ms | 20 |
| Duck Release | `mix.duck_release` | 10 .. 800 ms | 125 |
| Duck Release (Lines) | `mix.duck_release_lines` | 10 .. 800 ms | 65 |
| Counter under Lead | `mix.counter_duck` | 0 .. 1 | 0.2 |
| Pad Band under Lead | `mix.pad_lead_duck` | 0 .. 12 dB | 3 |

### GAIN

| Control | Key | Range | Default |
|---|---|---|---|
| Gain | `master.gain` | -24 .. 24 dB | 0 |
| Ceiling | `master.ceiling` | -12 .. 0 dBTP | -1 |
| Clip | `master.clip` | off / on | on |

### COMPRESSOR

| Control | Key | Range | Default |
|---|---|---|---|
| Comp Threshold | `master.comp_threshold` | -40 .. 0 dB | -8 |
| Comp Ratio | `master.comp_ratio` | 1 .. 10 :1 | 1.5 |
| Comp Knee | `master.comp_knee` | 0 .. 24 dB | 6 |
| Comp Attack | `master.comp_attack` | 0.1 .. 100 ms | 30 |
| Comp Release | `master.comp_release` | 10 .. 2000 ms | 200 |

### OUTPUT

| Control | Key | Range | Default |
|---|---|---|---|
| Mono Bass | `master.mono_bass` | 40 .. 300 Hz | 120 |
| Limiter | `master.limiter` | off / on | on |
| Limiter Release | `master.limiter_release` | 2 .. 1000 ms | 20 |
| Target Loudness | `master.target_lufs` | -20 .. -4 LUFS | -9 |
| Auto Gain | `master.auto_gain` | off / on | on |
| Clipper | `master.clipper` | off / on | on |
| Clipper Threshold | `master.clipper_threshold` | -6 .. 6 dB | 0 |

### MONITOR

| Control | Key | Range | Default |
|---|---|---|---|
| Monitor | `master.monitor` | Normal \| Mono \| Sub \| Side | Normal |

## 18. Perform

Four macros, meant for a hand rather than a mouse.

They are immediate parameter moves rather than score events, and that is measured rather than convenient: the engine's control ring is first in, first out and the composer keeps it filled eight bars ahead, so an event pushed into it now would not be heard for about thirteen seconds. A macro therefore writes the knob itself, as an offset in the knob's normalised domain, and puts it back exactly where it was when you let go. The composer's own ramps keep riding on top.

Drop-out is a press, not a hold: it takes the kick and the bass out and lets them back in on the next bar line. Stutter is held. Filter Sweep and Gate Depth stay where you leave them, and a double click puts them back to neutral.

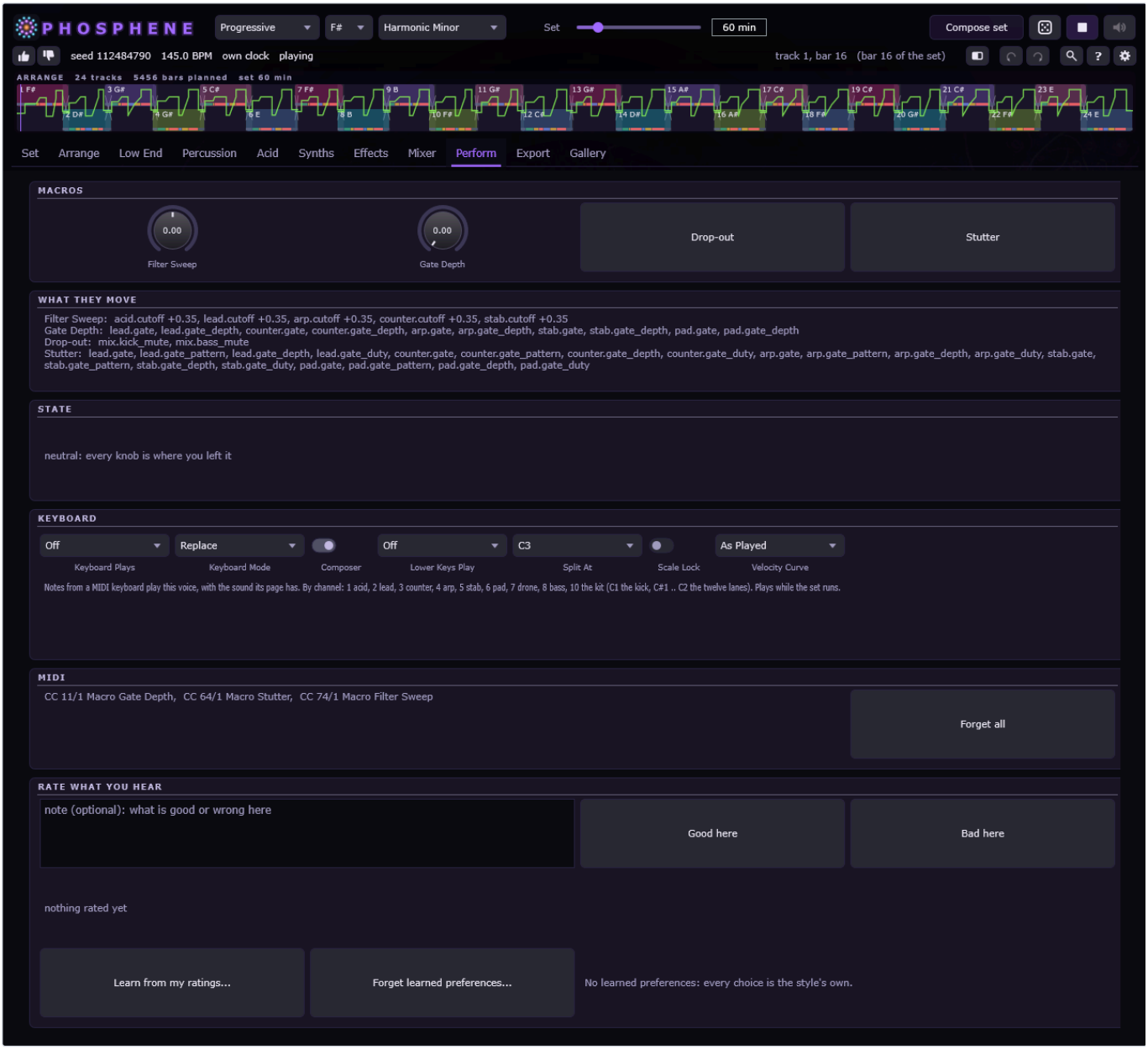
Keyboard: notes from a MIDI keyboard can play one voice -- the acid, any of the six melodic voices, the bass or the kit (C1 the kick, C#1 to C2 the twelve lanes), or "By channel" (channel 1 the acid, 2 the lead ... 7 the drone, 8 the bass, 10 the kit) -- with the sound that voice's tab has. Replace leaves the generator's notes of that voice out, Layer plays over them. Composer off leaves every generated note out: then only what you play sounds, through the mix and the effects as composed. The keyboard plays while the set runs; an export always plays what was composed.

Three options of the Keyboard group, each off until you choose it: Lower Keys Play splits the keyboard -- the keys below Split At (C1 to C5, C3 to begin with) play the voice chosen there, the keys from it upwards the voice of Keyboard Plays; Scale Lock moves a key outside the track's key and mode to the nearest note inside it, so no wrong note can sound; Velocity Curve shapes the touch -- As Played, Soft (a light touch is enough), Hard (it takes a firm one) or Fixed (every note at 100, for pads that send no velocity). A key's release always ends the note its press started, even when an option changed while it was held.

MIDI: every knob, switch and chooser, and the four macros, can be put on a hardware controller. Right click the control, choose MIDI Learn and turn the knob on your controller. The MIDI group lists every binding and forgets them all at once. Without learning, controller 74 is the filter sweep, 11 the gate depth and 64 the stutter -- the controllers every generator of the family listens to.

Headset: while a Meta Quest sends its hands (the Quest app's bridge mode), the Perform tab shows what they do -- the left pinch play and stop, the right pinch the drop-out, held the stutter, both hands the next track, the left hand the filter sweep, the right the gate depth (The panel).

Rate what you hear: Good here and Bad here write a verdict with a note into your ratings. Learn from my ratings fits your taste to them and, once you confirm, weights the choices the style leaves open towards what you liked. The genre's rules always stay above your taste.




The Perform tab.

### Filter Sweep

Opens or closes the filters of acid, lead, counter-lead, arp and stab together, by up to a third of their range in each direction. Centre is neutral.

*Moves:* `acid.cutoff +0.35, lead.cutoff +0.35, arp.cutoff +0.35, counter.cutoff +0.35, stab.cutoff +0.35`

### Gate Depth

Switches the trance gate on for lead, counter-lead, arp, stab and pad and sets its depth; at 1.0 the gate closes completely between its steps.

*Moves:* `lead.gate = 1, lead.gate_depth = 1, counter.gate = 1, counter.gate_depth = 1, arp.gate = 1, arp.gate_depth = 1, stab.gate = 1, stab.gate_depth = 1, pad.gate = 1, pad.gate_depth = 1`

### Drop-out (press)

Takes kick and bass out until the next bar line and lets them back in on the downbeat. One press, one bar.

*Moves:* `mix.kick_mute = 1, mix.bass_mute = 1`

### Stutter (press)

Held: lead, counter-lead, arp, stab and pad run through the gate on sixteenths at full depth and the shortest duty. Phosphene has no buffer repeat, so this is the gate's stutter, not a tape one.

*Moves:* `lead.gate = 1, lead.gate_pattern = 0, lead.gate_depth = 1, lead.gate_duty = 0, counter.gate = 1, counter.gate_pattern = 0, counter.gate_depth = 1, counter.gate_duty = 0, arp.gate = 1, arp.gate_pattern = 0, arp.gate_depth = 1, arp.gate_duty = 0, stab.gate = 1, stab.gate_pattern = 0, stab.gate_depth = 1, stab.gate_duty = 0, pad.gate = 1, pad.gate_pattern = 0, pad.gate_depth = 1, pad.gate_duty = 0`

## KEYBOARD

| Control | Key | Range | Default |
|---|---|---|---|
| Keyboard Plays | `mix.keyboard_part` | Off \| Acid \| Lead \| Counter \| Arp \| Stab \| Pad \| Drone \| By channel \| Bass \| Kit | Off |
| Keyboard Mode | `mix.keyboard_mode` | Replace \| Layer | Replace |
| Composer | `mix.composer` | off / on | on |
| Lower Keys Play | `mix.keyboard_lower` | Off \| Acid \| Lead \| Counter \| Arp \| Stab \| Pad \| Drone \| Bass \| Kit | Off |
| Split At | `mix.keyboard_split` | C1 \| C2 \| C3 \| C4 \| C5 | C3 |
| Scale Lock | `mix.keyboard_scale` | off / on | off |
| Velocity Curve | `mix.keyboard_velocity` | As Played \| Soft \| Hard \| Fixed | As Played |

## 19. Export

What leaves the instrument, in one place, as every generator of the family has it (01.10.2026): Record (the red disc) writes the output as it plays into a WAV file until it is pressed again; Export MIDI writes the score of so many bars from the start, a channel per part; the disk writes the set -- the seed, the locks, the rerolls and every knob that differs from the factory -- as a .phosset, and the folder reads one and plays it. The score cues (OSC) send the set's beats, bars, sections and keys to a light show or a VJ program at the moment they are heard.

The keys: Ctrl+E exports the MIDI, Ctrl+S the set, Ctrl+O loads one.

Four things connect Phosphene to the software around it and to the rest of the family (02.10.2026). The outputs, Link and the jam are off until they are wanted, the MIDI goes nowhere until a track takes it, and only the jam -- as a follower -- changes what the music sounds like.

MIDI out: in a DAW the plugin also sends what it plays -- every note of every part at the moment it sounds, a channel per part as in the MIDI file of an export -- and leaves out a voice the keyboard plays instead (Replace). Record it on a MIDI track to keep a part as notes, or route it to another instrument to double it. A stop or a jump of the transport sends all notes off.

Outputs of their own: besides the main output the plugin has a stereo output for each of the stems -- the kick, the bass, the percussion, the acid, the lead, the counter, the arp, the stab, the pad, the drone, the effects, the texture, the voice, the field recordings and the returns. They are off to begin with; switched on in the DAW (in most hosts: the plugin's output configuration, or its multi-output version), each carries its part before the master, while the main output plays on as before.

Ableton Link (the standalone; Settings > Ableton Link): with another Link app on the network -- Live, Bitwig, a phone app, another generator of the family -- Phosphene takes the session's tempo, lines its bars up with the session's and starts and stops with it; alone in a session it offers its own tempo to whoever joins. In a DAW the host's transport leads, as before.

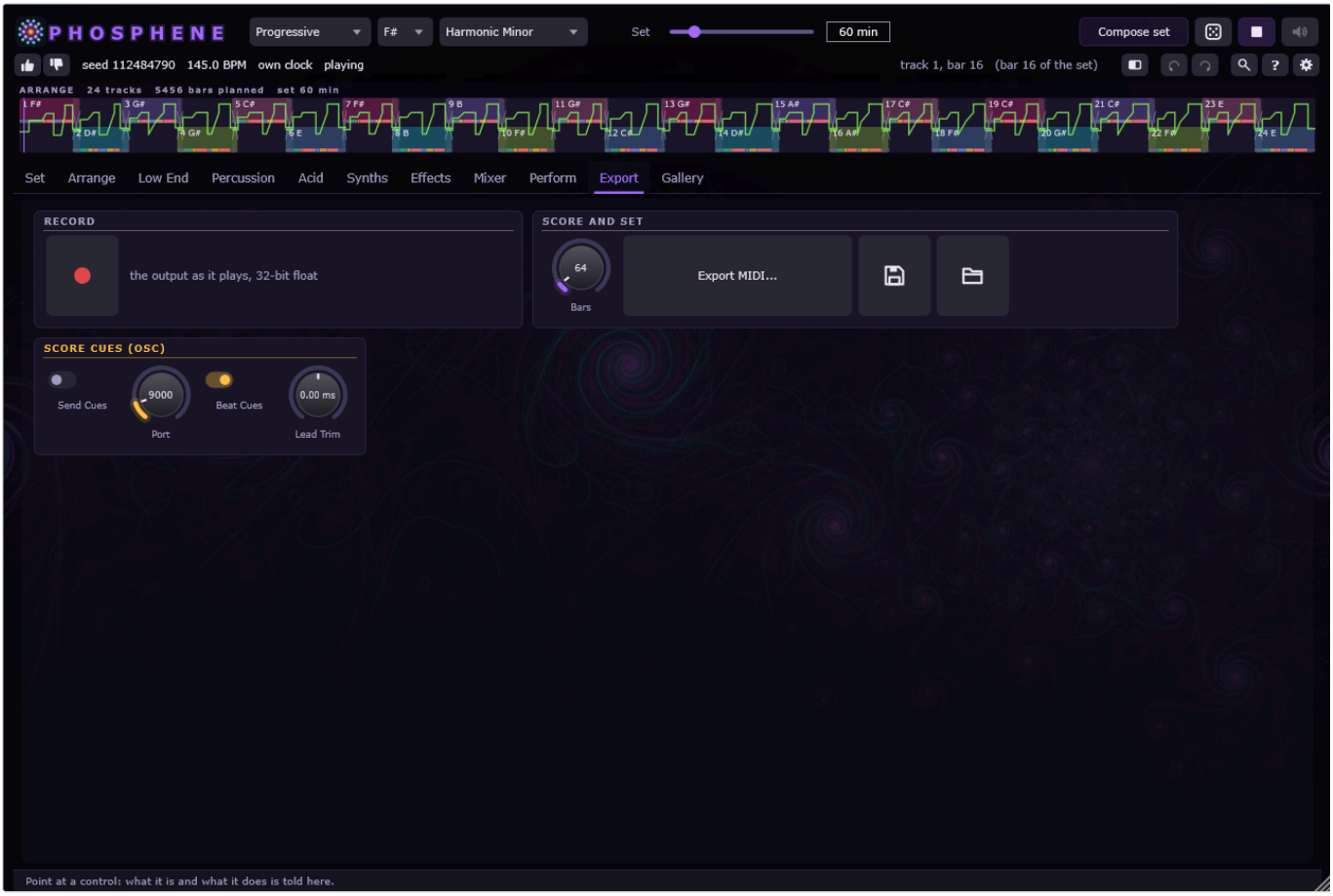
The family jam (Settings > Family jam, in the plugin and in the standalone; off to begin with): the five instruments of the family -- Totality, Parhelion, Ephemeris, Phosphene and Noctuary -- play as one band on the local network. One leads: its key, the energy of its sections, its breaks and its drops go out to the others. Those set to Follow take them up -- in Phosphene: the bass, the 303, the lead, the counter, the arp, the stab, the pad and the drone move to the leader's key while the kit and the effects play as written, and the set's walk holds its key; the Filter Sweep macro closes with a quiet section of the leader's; and in the leader's breaks the kick and the bass are out until its drop. A major mode of the leader's it leaves be: psytrance's palette has none. A new root comes in at the follower's next bar line, by the shortest way, the mode with its next track. Tempo and bars come from Ableton Link in the standalone (switch it on in each of them) or from the DAW's transport, so that a section the leader announces lands on the same bar everywhere. As the leader Phosphene says its set's key and each section of the set as it passes, with its energy. The jam stays on this network (UDP multicast to 239.255.77.77, port 9177, not past the router), instruments on one machine find each other as well, and a follower that has not heard its leader for four seconds goes back to its own key and its own form. The last line of the submenu says what it does: off, leading, or whom it follows.

On a Mac: every release has a build for Apple Silicon (macOS 12 or newer), the standalone, the VST3 and the Audio Unit (Logic, GarageBand, MainStage; passed by Apple's auval), built and tested on GitHub's machines. It is signed ad hoc, not notarized: macOS asks once (README-macOS.txt in the zip says how), and it has not yet been played on a real Mac -- reports are welcome.

On Linux: every release has an archive for x86-64 (glibc 2.35 or newer: Ubuntu 22.04, Debian 12, Fedora 36 and later) with the standalone, the VST3, the LV2 (Ardour, Carla, Reaper, Qtractor) and the renderer, built and tested on GitHub's Ubuntu machines and tried under WSL. README-Linux.txt in it says where everything goes: the VST3 into ~/.vst3, the LV2 into ~/.lv2, the standalone started where it lies, sound through ALSA or JACK (PipeWire serves both).

Demos: the release "demos" holds a track per style as an MP3 and one of them as a video, with pictures by KaleidoscopeEnhanced cut to the track's own score cues; Tools/demo/make_demos.py makes them all again (phos_render).

KaleidoscopeEnhanced understands the score cues of all five instruments as they come (since 02.10.2026) -- Totality's blocks and keys, Parhelion's sections, Ephemeris' phases, Phosphene's sections and drops, Noctuary's bars, keys and scenes -- and cuts its pictures to them, on a drop at once.



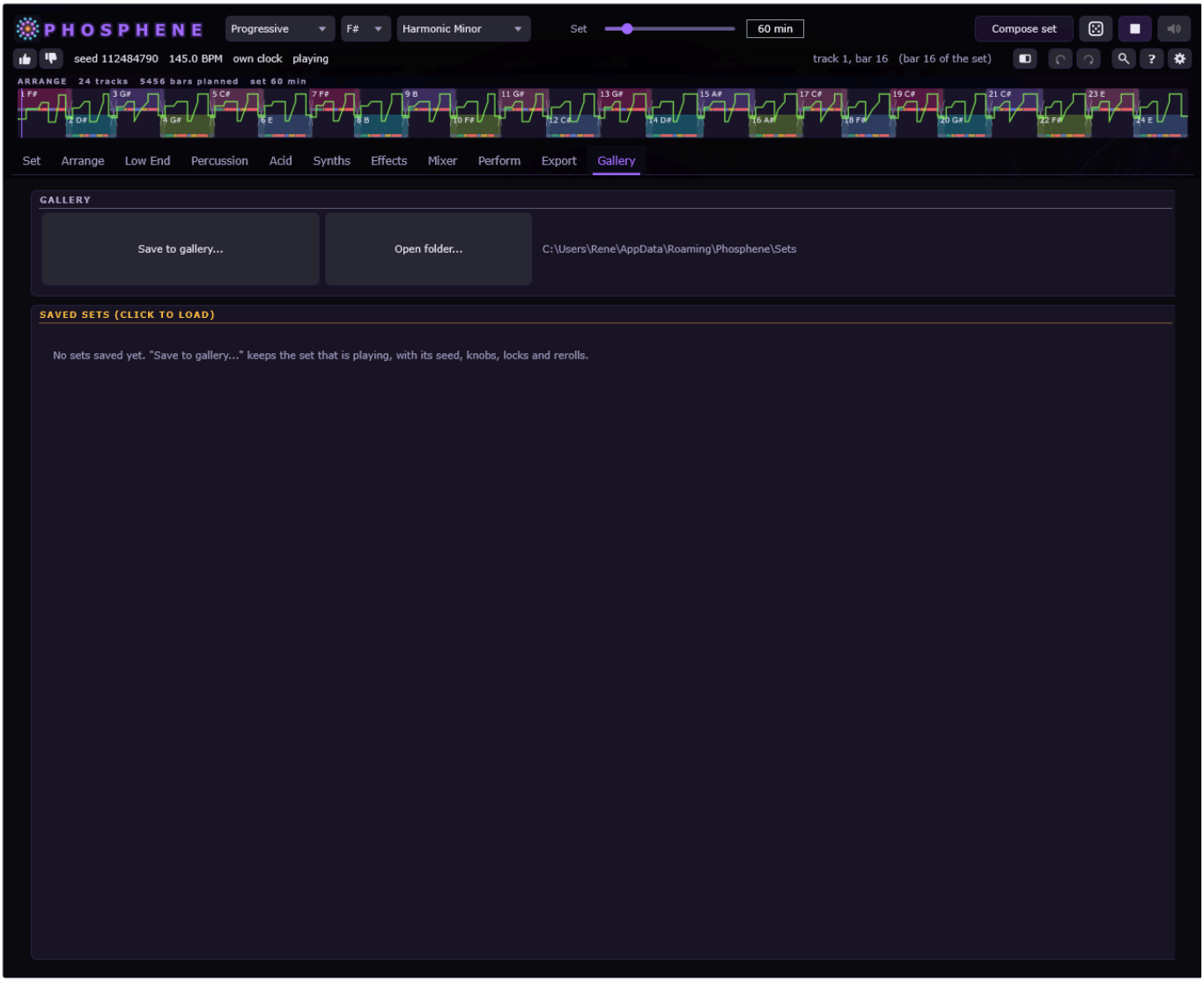

The Export tab.

## SCORE CUES (OSC)

| Control | Key | Range | Default |
|---|---|---|---|
| Send Cues | `cue.send` | off / on | off |
| Port | `cue.port` | 1024 .. 65535 | 9000 |
| Beat Cues | `cue.beats` | off / on | on |
| Lead Trim | `cue.lead_ms` | -50 .. 50 ms | 0 |

## 20. Gallery

Your saved sets, each with a picture of its form. Save to gallery keeps the set as it stands -- seed, style, arc, knobs, locks and rerolls -- and a click on an entry loads it back. The files are ordinary .phosset files in your Sets folder; Open folder shows it.




The Gallery tab.

## 21. Filters and modulation

The melodic voices and the bass choose their filter's circuit with Filter Model. The first entry is the synth's own filter -- the voices' state-variable filter, the bass's ladder -- and the others are circuit models solved sample by sample: the Moog transistor ladder, the OTA cascades of the Prophet and the Juno, the Oberheim SEM, the Xpander's pole mixes, the diode ladder of the TB-303, the Korg35 of the MS-20, the Polivoks and the Wasp. Filter Mode morphs the SEM from low pass through notch to high pass, steps through the Xpander's eight responses and turns the Polivoks into a band pass. Switching models keeps the level: each is calibrated against the synth's own filter. The acid keeps its diode ladder -- that is the acid.

The filter envelope is a full ADSR: Filter Attack, Decay, Sustain and Release. With the attack at its minimum and the sustain at 0 it is the plain decay every older set used.

A third envelope (Mod Envelope), four LFOs and a modulation matrix of eight slots move the rest. An LFO runs free at its rate or synced to the beat (4 bars down to a sixteenth, and triplets), in seven shapes from sine to smooth random; Retrig restarts it with every note, Fade brings it in over a few seconds. A slot of the matrix reads a source -- an LFO, the mod or filter envelope, the velocity, the key or a random value drawn per note -- and moves a target by its amount: pitch, the second oscillator's pitch, pulse width, the wavetable position (the classic slow scan through a table), the FM index, cutoff, resonance, the filter mode, the level or the pan. The bass offers only what cannot break its lock to the kick (no pitch, no pan), the acid what suits a 303 line.

## 22. Sound presets

Every synth tab starts with a Sound group: the synth's presets in a chooser, grouped by kind, a disk for presets of your own, and the Own Sound switch.

Every track of a set plays a preset on every synth that is not on Own Sound: the composer chooses one by the track's style -- a Hoover or a Supersaw Anthem for Full-On, a Growl or a Screaming Glide for Dark Forest, a Glass Lead or a Psy Pluck for Progressive -- and never the same group as the two tracks before. While it plays, the tab shows it: the chooser reads "Group: Name (composer)" and the knobs stand at the preset's values. The sections still move some of them on top, the way a player rides a filter; those moves are not written into the knobs. The composer also evens out the presets' loudness: each carries a correction, measured against the synth's own default sound, that rides on the synth's Level on top of your setting, and the level match then holds every part of every track -- the first one included -- to the loudness the default sound would have there. Sound Presets on the Set page (or Sound Variation at 0) turns the composer's presets off, and every synth plays its knobs again, with the composer's recipes.

Own Sound keeps the composer out: the synth then sounds exactly as its tab says, in every track. Turning a knob of a synth that plays the composer's preset switches its Own Sound on, with the knobs where the preset put them, so the sound you start from is the one you heard. Choosing a preset from the chooser switches Own Sound on as well. A preset sets the synth's sound and leaves its level, its ducking, its sends to the rooms, its place in the image and -- on the melodic voices -- its high pass and trance gate alone, since those are where the mix and the form put the voice. Undo takes a preset back.

The live ring: where a knob plays away from where it stands -- a section riding the cutoff, drop 2's lift, a preset's level correction -- a thin bright arc on the knob's ring runs from the knob to the value that plays, with a dot where it plays. The modulation shows there too: an LFO or the mod envelope on the cutoff, the table position, the resonance, the filter mode, the pulse width, the FM index, the level or the pan moves the dot as the latest note's voice moves. A knob without it plays exactly what it shows.

The factory bank holds 1024 presets for each synth: sixteen groups -- Hoover, Acid Lead, Vocal Formant, Rolling, Squelchy, Classic 303, Full-On ... -- of sixty-four each. A group is a grid of eight adjectives from dark to bright (the lead's run from Umbral to Supernova) and eight nouns of its own (the Hoover's from Hoover to Titan), so "Umbral Mentasm" is the darkest of the Hoovers' second column and "Supernova Titan" the brightest of its last. Each preset brings the filter model its group suits and modulation from its group's recipes: a slow LFO on the cutoff, the wavetable scanned by an LFO or the mod envelope, a hoover's pitch dive, a vibrato that fades in. Your own presets are listed under User and live in the Presets folder of your Phosphene user folder.

## 23. The panel

Phosphene shares its panel with the other generators of the family -- Ephemeris, Totality and Parhelion (and, in part, Noctuary) -- so that what is learned on one holds on the others: the same places, the same keys, the same controllers, the same hands. Each keeps its own colours, its own letters and the picture behind its panel (Settings > Picture behind the panel takes it away).

The header's first row: the logo, the style, the key and the scale, what is composed and its length, Compose, the die (a new seed), Play and the speaker (mute). Its second row: Good here and Bad here -- a verdict on what plays, for your taste --, the status line, and at the right Undo, Redo, Help and the settings. Under the header the whole set at a glance; a click jumps there. The tabs come in one order everywhere: Set, Arrange, the instrument's own pages -- a page of several has small tabs of its own --, Mixer, Perform, Export, Gallery.

The unambiguous actions are icons since 03.10.2026, their words in the tooltips: the die draws a new seed, the triangle plays and the square stops, the speaker mutes (crossed out while muted), the disk saves and the folder loads, the red disc records (Export).

The ways around the panel (15.10.2026, the family's). The magnifier at the right of the header (Ctrl+K) opens the search: a few letters find any knob -- its page opens, its section or lane comes to the front and it lights up; a number after its name sets it ("cutoff 800") --, any preset of any synth, a page or an action. The line along the bottom of the window says what the control under the mouse is: its value now, its range and default, where the score plays it, what moves it, its MIDI controller and its key. Each synth's Sound group has a magnifier, the browser: by name, by group, the starred and the recently played; hold a row's triangle to hear it, double-click to take it. The name under the chooser carries a star while the knobs have moved from the preset chosen. A knob a voice's modulation moves wears an arc round it for every row of the matrix on it -- the span the row can move it over, in its source's colour; a right click lists those rows, each with its depth to drag and a button to take it out, and offers every source (Modulate with: a row in the first free slot, a quarter of the target's span). Held, the split square in the header (A|B) plays the presets as they were chosen; let go, the sound as it has been turned since. The first steps show the panel on the first start (Settings > First steps, again). The standalone's window comes back where it was left -- its place, its size, maximised or not --, and its audio settings offer ASIO (Windows).

A page taller than the window shows its groups in sections, one at a time, with a switch at its top right: a synth's sound and its modulation apart (Sound, Modulation), and where even those do not fit, cut further, each named after its first group (Filter, LFO, Matrix ...). At the window's usual size, 1280 by 860, no page scrolls. The overview above the tabs can be folded away in the settings; the pages then get its height.

A colour says what a control does, the same on every page and in every generator of the family: the sources, the filters, the envelopes, what moves by itself, the room and the mix. A knob the composer moves has a thin ring round it that shows where it stands at this moment. A double click sets a control to its default; a right click offers MIDI learn, forgetting the controller, and the default.

Undo and Redo take back and bring back what was changed -- a knob, a new seed, a reroll, a preset, a loaded set -- one step at a time; the button's tooltip names the step. Help opens this manual by topic, with the page in front, the keys and the headset as topics of their own. The settings hold the update check (it asks GitHub once a day whether a newer Phosphene is out; nothing else is sent), the picture behind the panel, the overview, the headset, the window's size, full screen, the keys and About.

```
Space                 play / stop
Ctrl+Z                undo
Ctrl+Y, Ctrl+Shift+Z  redo
F1                    help
F11                   full screen (the standalone); Esc leaves it and closes the help
Ctrl+S, Ctrl+O        save the set, open one
Ctrl+E                export
Ctrl+K                search: any control, preset, page or action
```

The controllers every generator of the family listens to without learning: controller 74 (brightness) is the filter sweep, 11 (the expression pedal) the gate depth, 64 (the sustain pedal) the stutter while it is held; the mod wheel is left free. A set with bindings of its own keeps them.

The headset. The controls of a Meta Quest's hands are shown only while a headset sends them -- or always, if the settings say so (Headset > Always show them); Off stops listening. The Quest app sends its hands to this computer in bridge mode: bridge_host = this computer's address in its phos.cfg (audio=0 leaves the headset silent), and Phosphene listens on port 9101. The hands mean the same in every generator of the family, on the headset alone and here:

```
left pinch            play / stop
both hands pinched    the next track
right pinch           drop-out: kick and bass gone until the next bar line
right pinch held      stutter, while it is held
left hand up, down    filter sweep: closed below the middle, open above
right hand up         gate depth
```

A pinch acts when it opens again, so a pinch of both hands is never also one of each; a hand moves its control only while it does not pinch, and both glide and are centred: a hand at mid height plays what was composed.

## 24. Shortcuts

```
Space            play / stop
F1               help (this page); Esc closes it
F11              full screen (standalone)
Ctrl+Z           undo
Ctrl+Y           redo (also Ctrl+Shift+Z)
Ctrl+S, Ctrl+O   export the set, load one
Ctrl+E           export MIDI
Ctrl+K           search: any control, preset, page or action
right click      MIDI Learn / Forget MIDI / Default value on any control
double click     a knob back to its default
```

The window can be dragged to any size; the whole panel zooms with it. The standalone's title bar has a maximise button, and Full screen in the settings (or F11) fills the screen.

## 25. Updates

Phosphene looks for a newer release at most once a day, when its window opens, by asking GitHub for the latest release of the project -- nothing else is sent, and nothing is installed by itself. When there is one, the header says so and a click opens the release page. The check can be switched off here on the help page or in the settings, and Check now asks at once.

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